

# Spatial diffusion in networks and churn of innovation

Lengyel Balázs

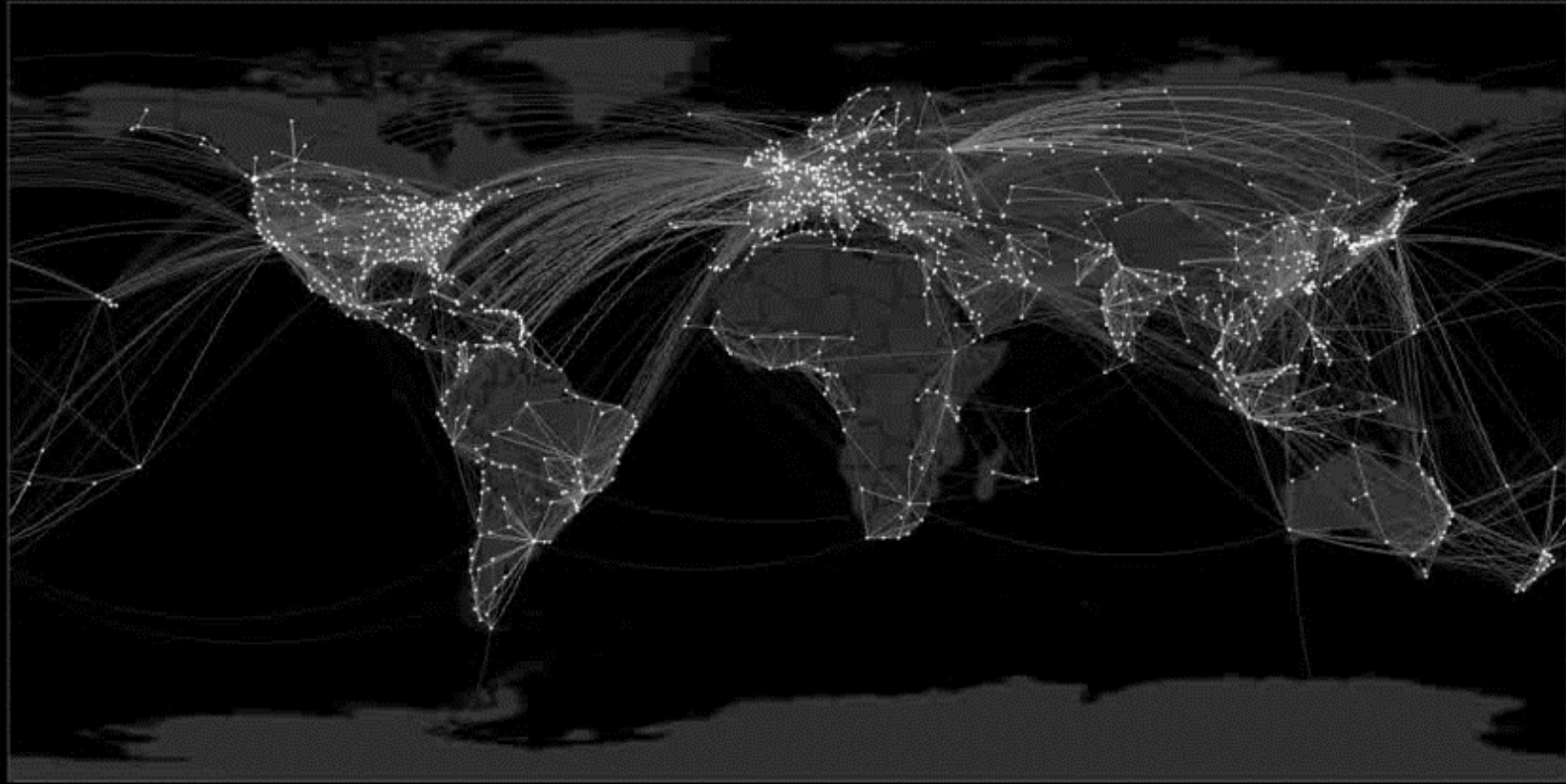
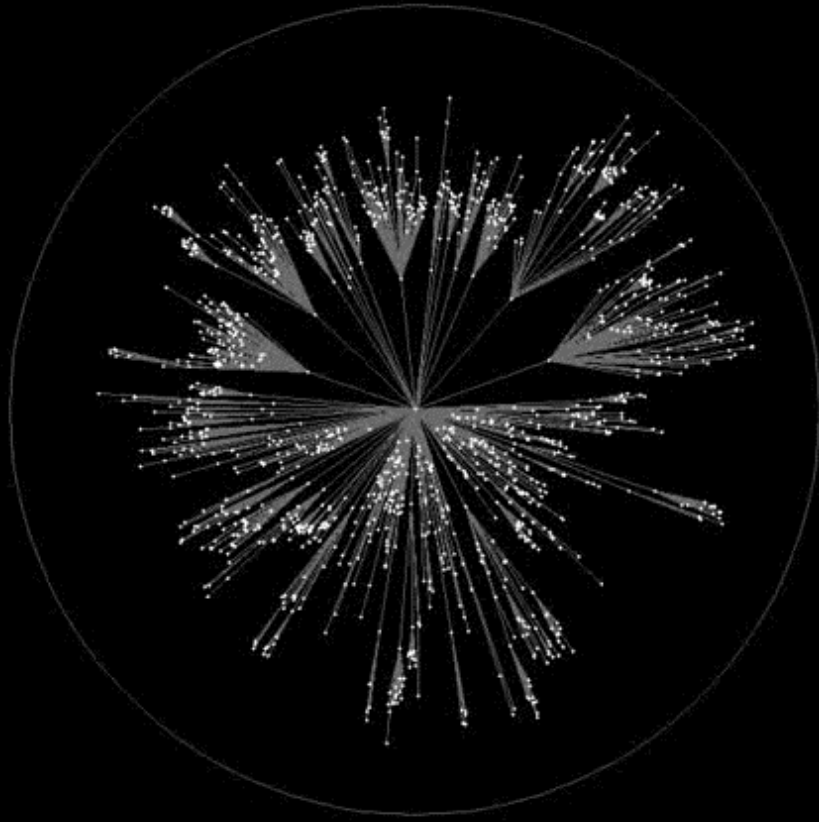
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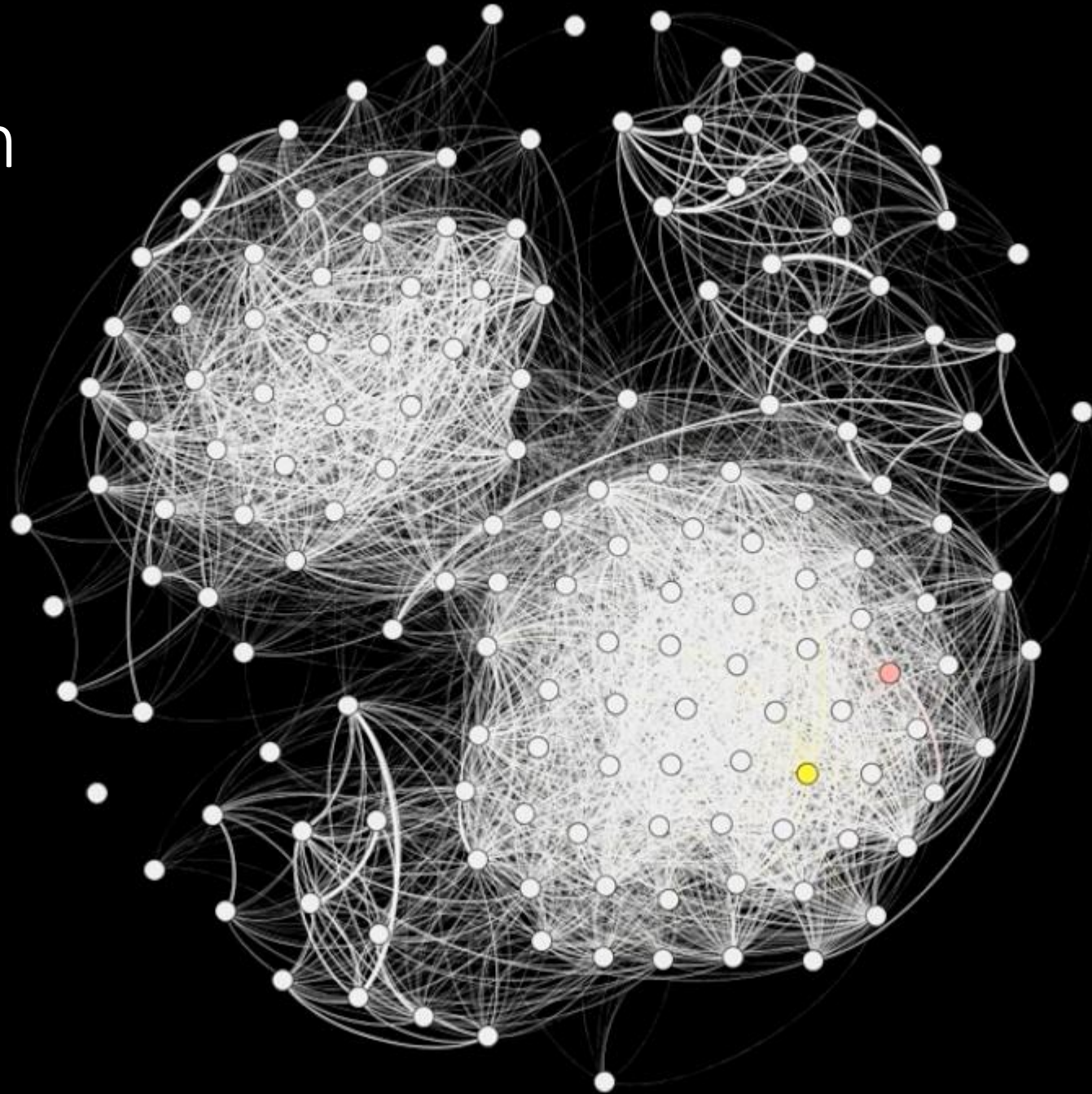
2026. May 8.

# Virus diffusion on the global flight network



Brockman D., Helbing D (2013) The hidden geometry in complex, network-driven contagion phenomena. Science 342 (6164) pp. 1337-1342.

# Diffusion in networks



# Today

1. Diffusion of innovation in networks
2. Spatial diffusion of innovation in networks
3. Spatial churn

# 1. Diffusion of innovation in networks

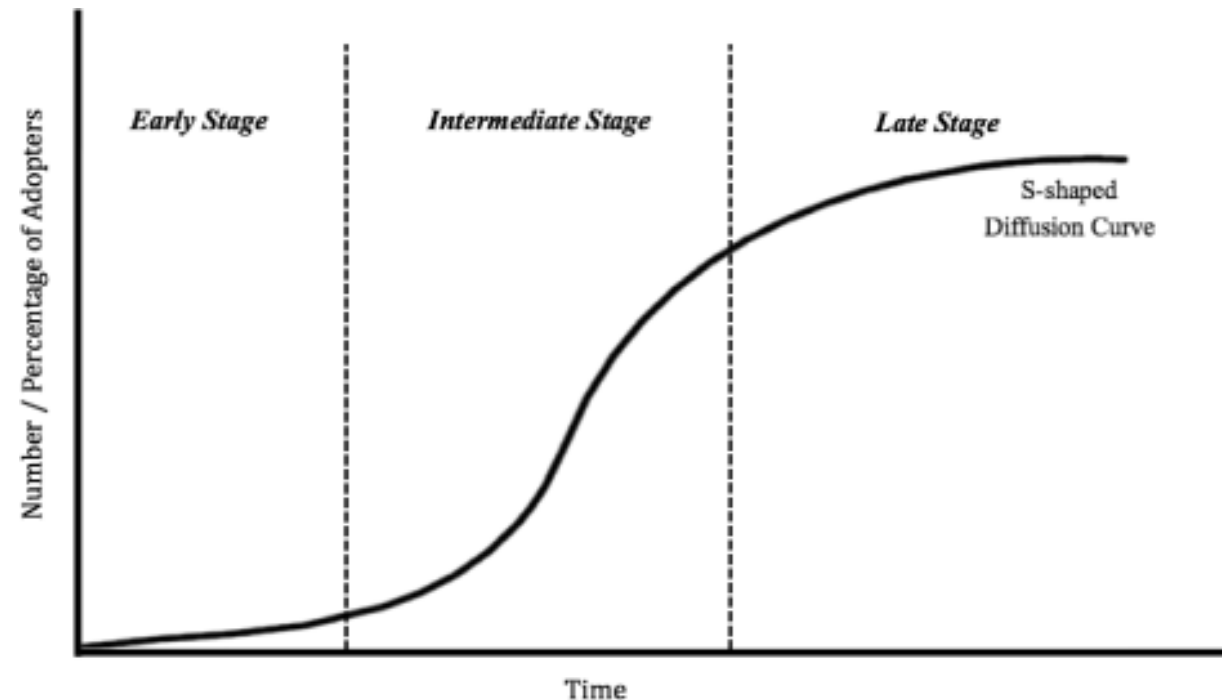
# The S curve of diffusion

Coleman, Katz, Menzel (1966): how fast do doctors prescribe doctors new medicines?

- diffusion: prescriptions
- network: survey information about advice links between doctors
- High degree doctors started to prescribe earlier.

Griliches hybrid corn example: how big fraction of farmers use it?

- Initially slow diffusion accelerates and slows down again.



# Bass model

Two states / activity: 0 or 1 – yes or no

- $F(t)$  – fraction of the population who started the activity until time  $t$ .
- $p$  is the rate of spontaneous adoption
- $q$  is the rate of imitation

$$\frac{dF(t)}{dt} = (p + qF(t)) (1 - F(t))$$

- Second term: fraction of population who have not adopted until time  $t$ .
- They can adopt spontaneously or follow the adopters.
- Solution to differential equation is:

$$F(t) = \frac{1 - e^{-(p+q)t}}{1 + \frac{q}{p}e^{-(p+q)t}}$$

# Bass model

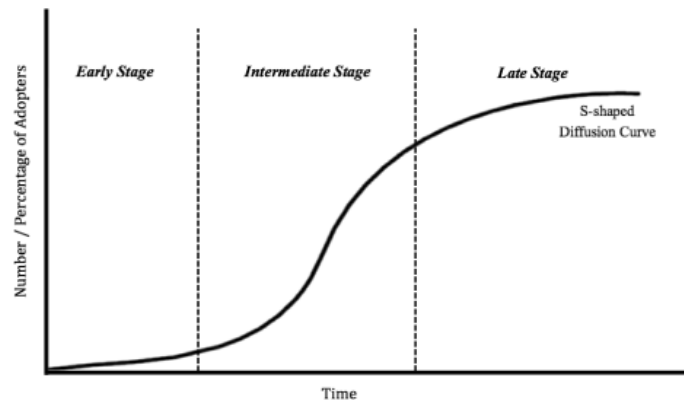
If  $F(t)$  is close to 1, diffusion slows down

- Second term goes to 0, growth goes to 0

$$\frac{dF(t)}{dt} = (p + qF(t))(1 - F(t))$$

In the beginning, only  $p$  matters, but later  $q$  becomes more important

- In the beginning  $F(t)$  is small, thus  $qF(t)$  goes to 0
- There is no one to adopt but anyone can adapt spontaneously
- The slope of spread is  $p$  in the beginning, but then  $q$  speeds this up.





# Bass model

The diffusion curve has S shape if  $q > p$ , that saturates at 1

If  $F(t)$  is close to 1, then  $d(F)/d(t) = 0$

If  $F(t) = 0$ , then  $d(F)/d(t) = p$

If  $F(t) = \varepsilon$ , then  $d(F)/d(t) = (p + q \varepsilon) (1 - \varepsilon)$

Convexity will be realized in the case:

$$(p + q \varepsilon) (1 - \varepsilon) > p$$

$$(q) (1 - \varepsilon) > p$$

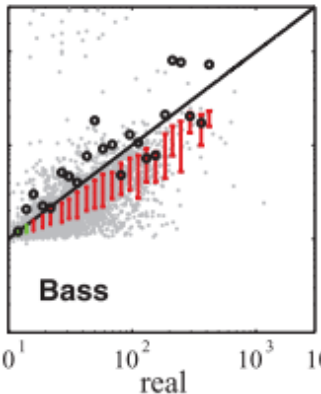
$$q > p$$

Usage:

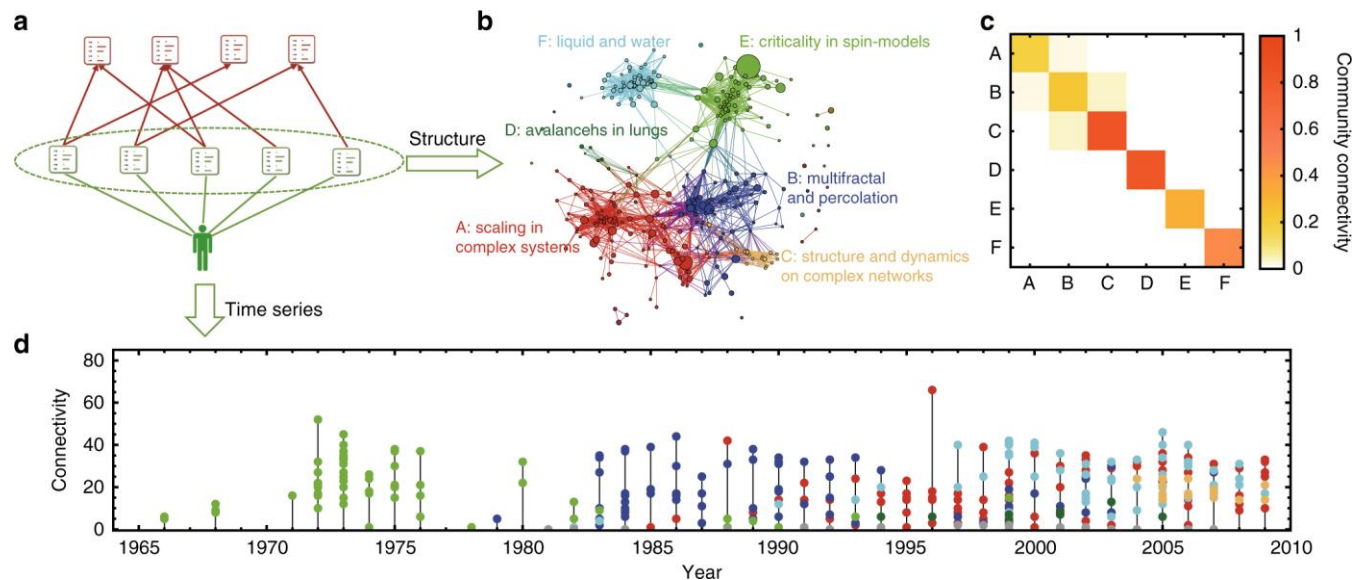
$p$  and  $q$  can be estimated at small  $\varepsilon$  and then predictions can be made

# Bass model research to predict impact (the LI70 hypothesis)

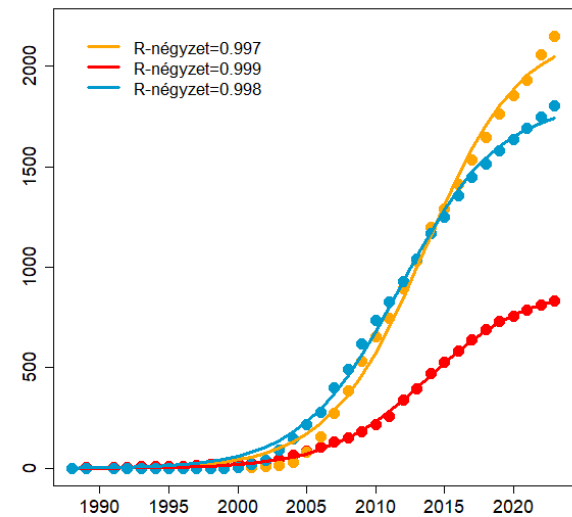
p and q can be estimated early and then predictions can be made



Wang et al. 2013 *Science*



Zeng et al. 2019 *Nature Communications*



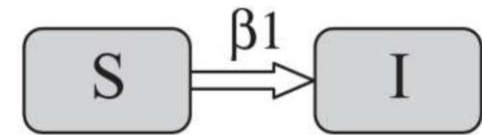
**LI70 hypothesis:** „Changes in research topics along the academic career make the prediction of success more difficult.”

# Bass vs SI models 1: random adoption

SI:

- N: size of population
- S: susceptible population. Those who can be potentially infected.
- I: infected population. Those who can infect others.
- $S(t)$ : susceptibles at time  $t$ .
- $I(t)$ : infected at time  $t$ .
- $\beta$ : probability of transmission

$$N = S(t) + I(t)$$

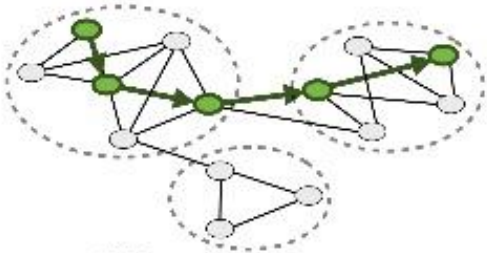


Bass:

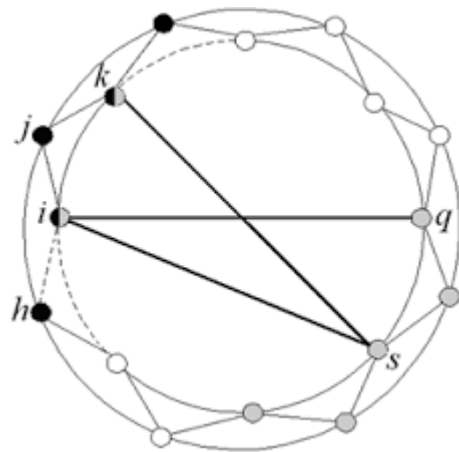
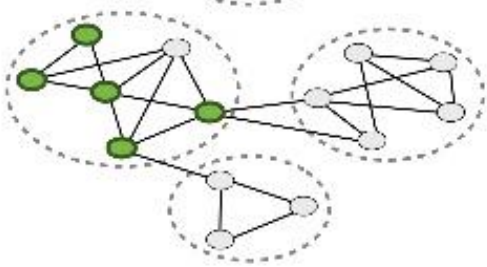
- Random adoption

# Bass vs SI: simple vs. complex contagion

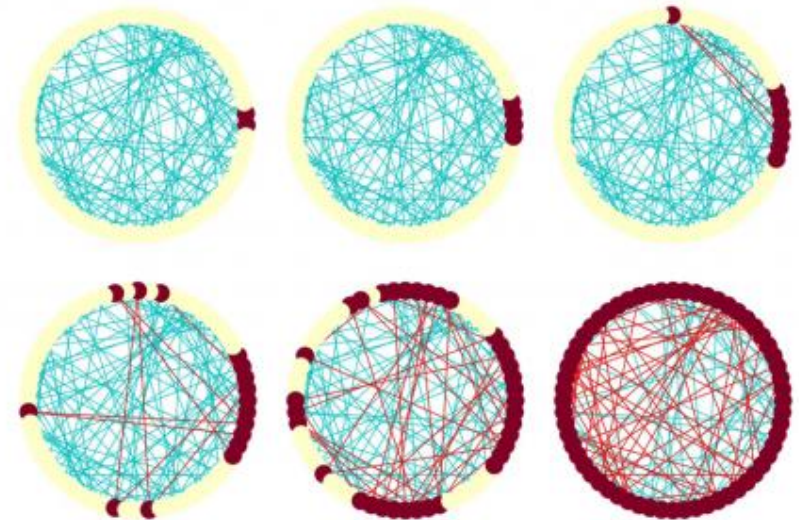
Simple contagion:  
weak concentration



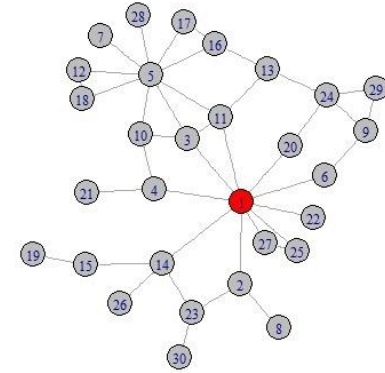
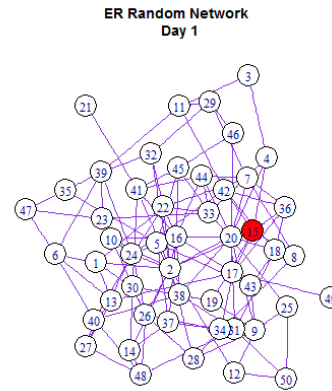
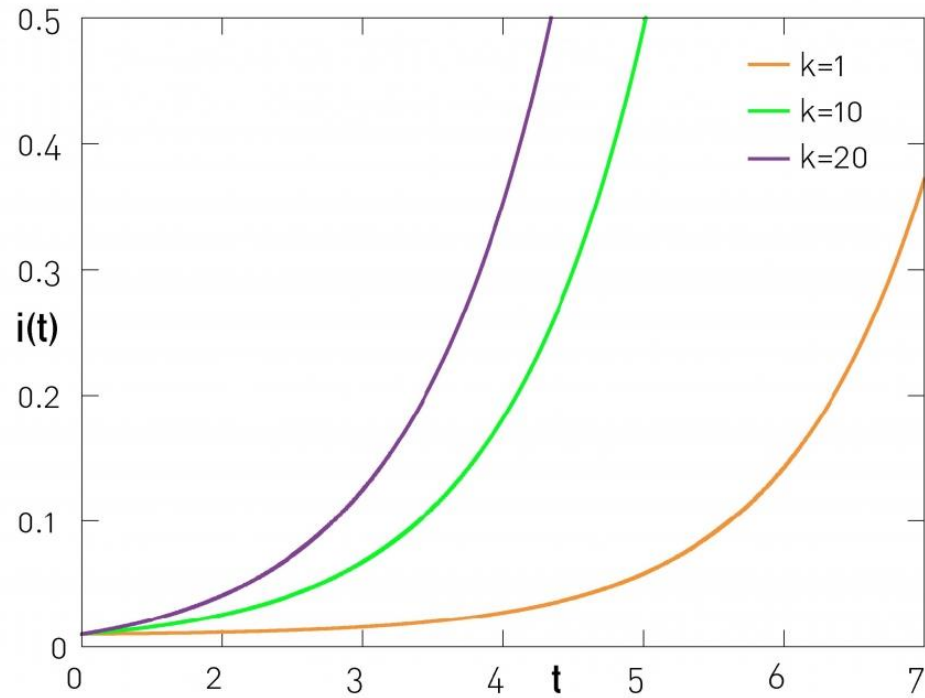
Complex contagion:  
strong concentration of  
communication inside  
communities



- Simple: infection depends on dyadic probability
- Complex (Centola and Macy, 2007): adoption is a process of convincing-decision and depends on fraction/number of adopting friends
- Innovation diffuses through complex contagion.
- Small-world networks: high clustering speeds diffusion up, bridges slow it down.

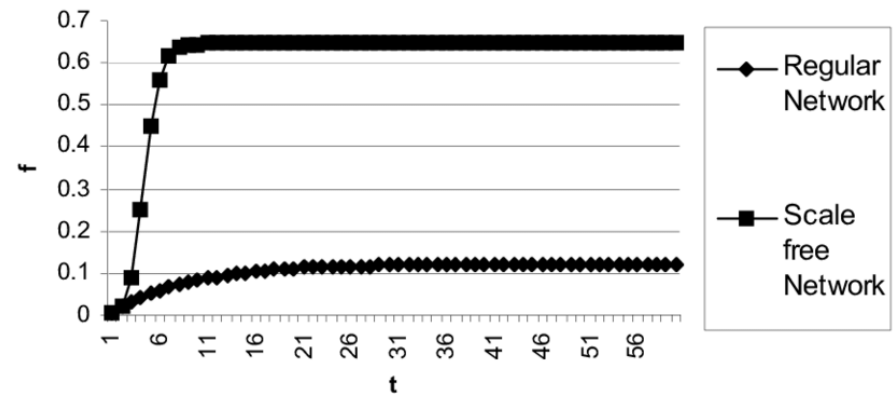


# Degree is key in SI



MakeAGIF.com

Speed of diffusion



# Threshold models of adoption

## Shelling: segregation in cities

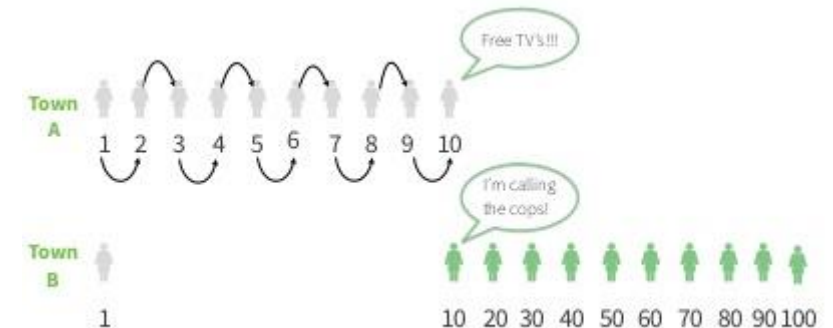
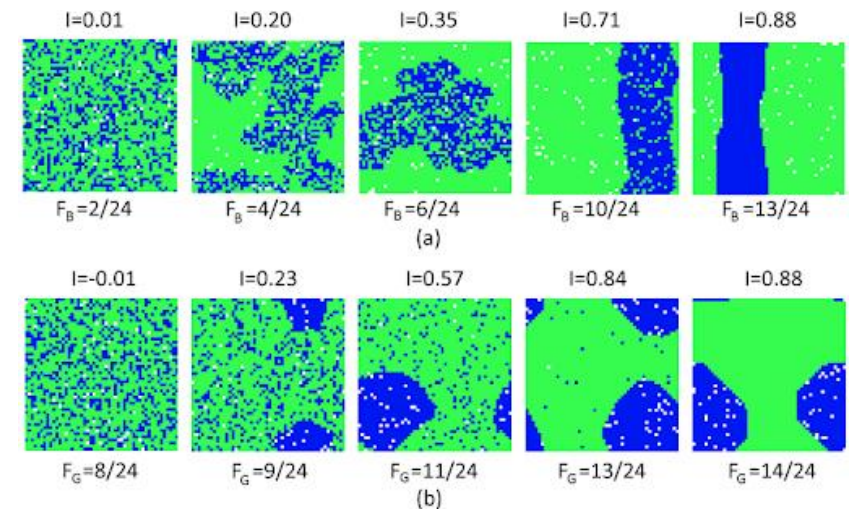
- Individual threshold in terms of ratio of tolerated different neighbors is heterogenous
- Low threshold individuals move away first that increases the probability of movement of high threshold individuals

## Granovetter: behavior in a protest

- Individual threshold: how many individuals is needed to start the activity before the individual will decide to start

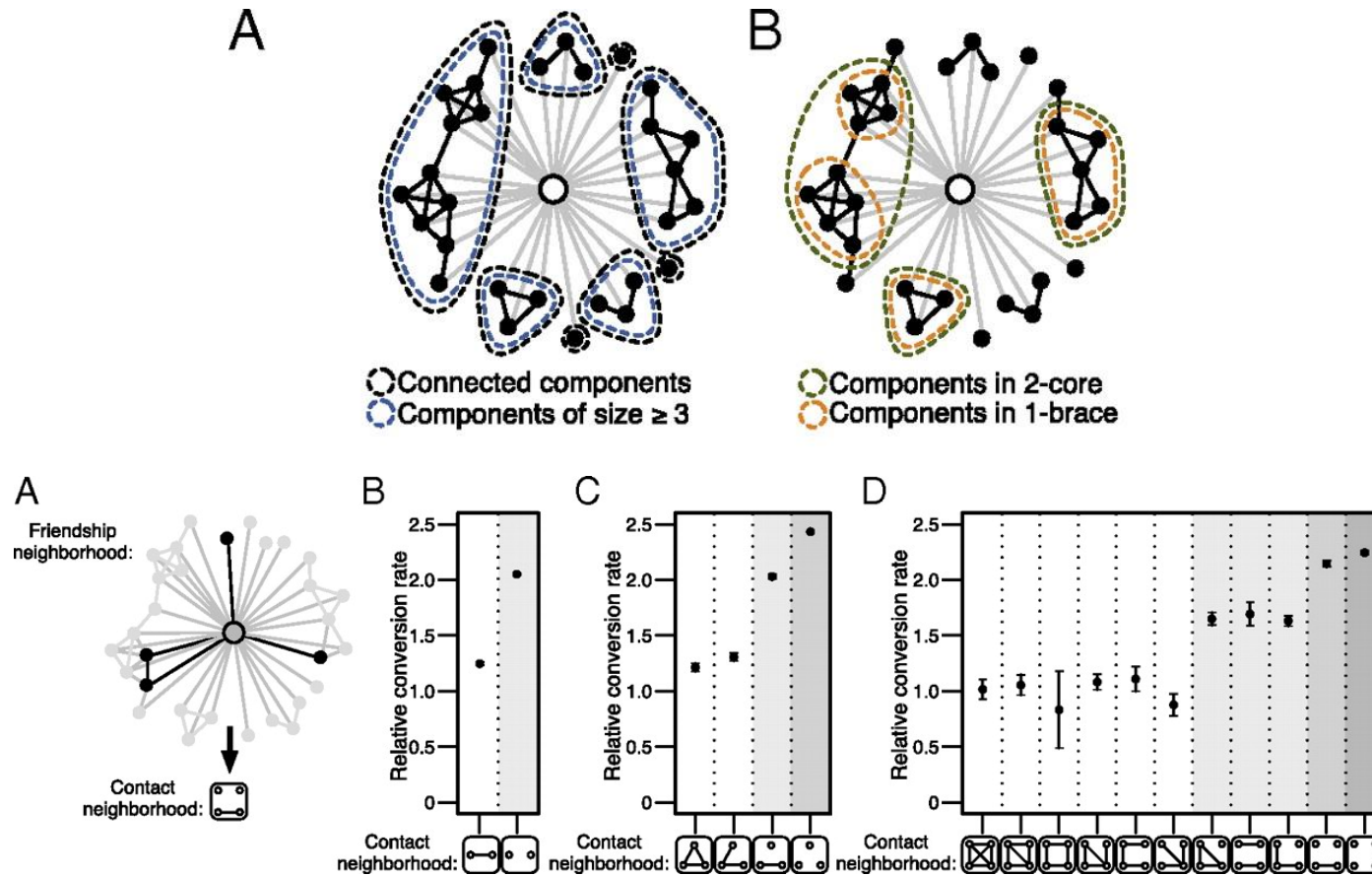
## Watts: cascading behavior in networks

- A behavior will diffuse only if the network structure follows the threshold sequence
- Innovators are connected to Early Adopters





# Role of communities: adoption probability increases after adoption in diverse communities

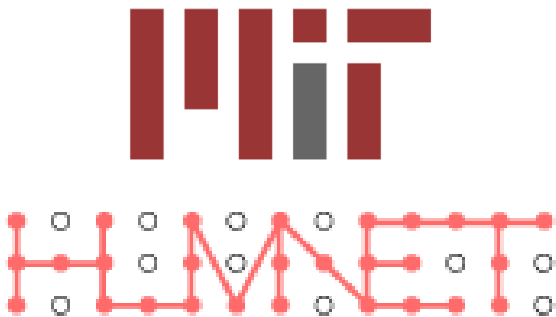


Ugander et al. 2012 PNAS

# The role of geography in the complex diffusion of innovations

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Riccardo di Clemente UCL



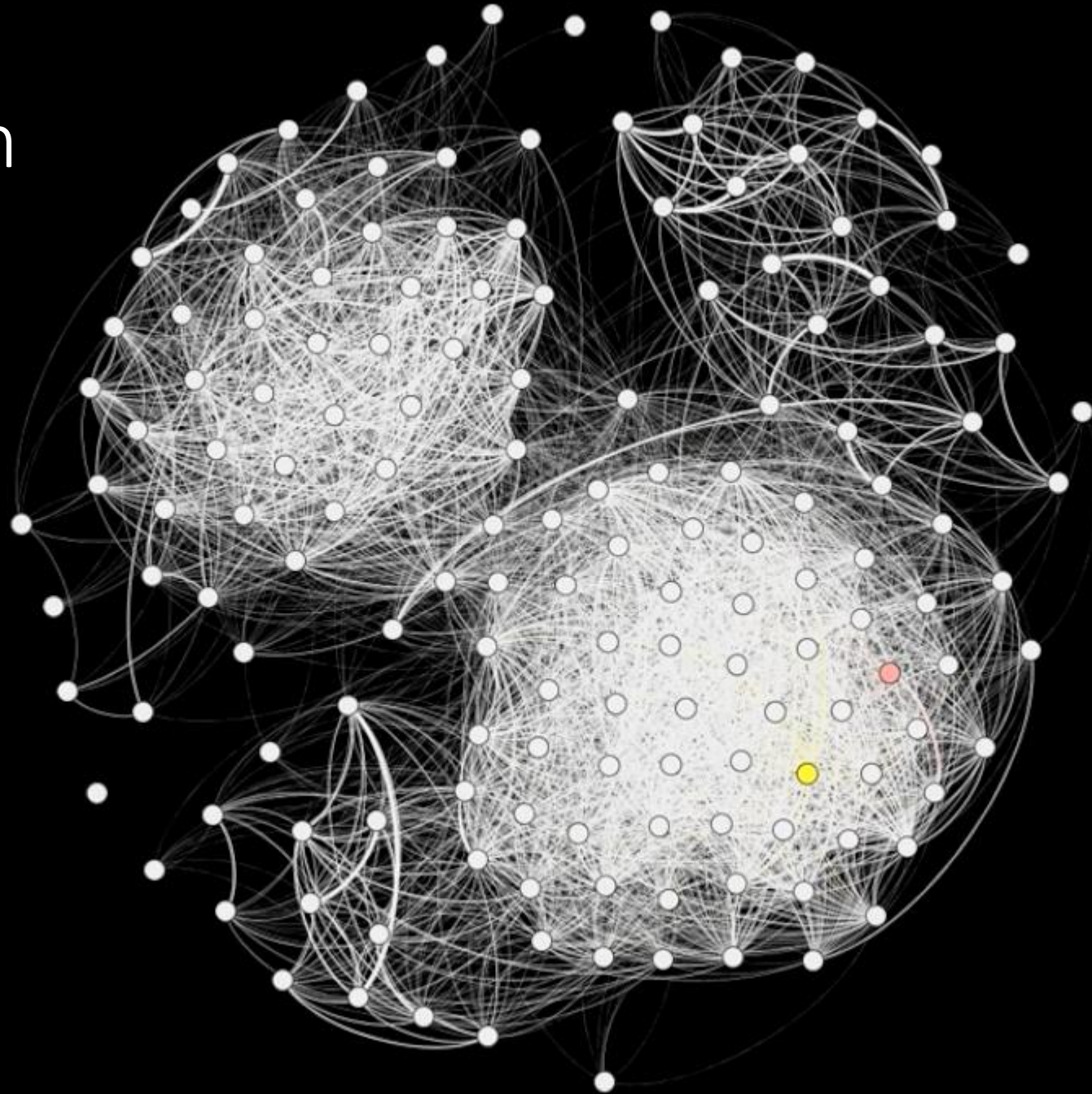
János Kertész CEU



Marta González  
UC Berkeley

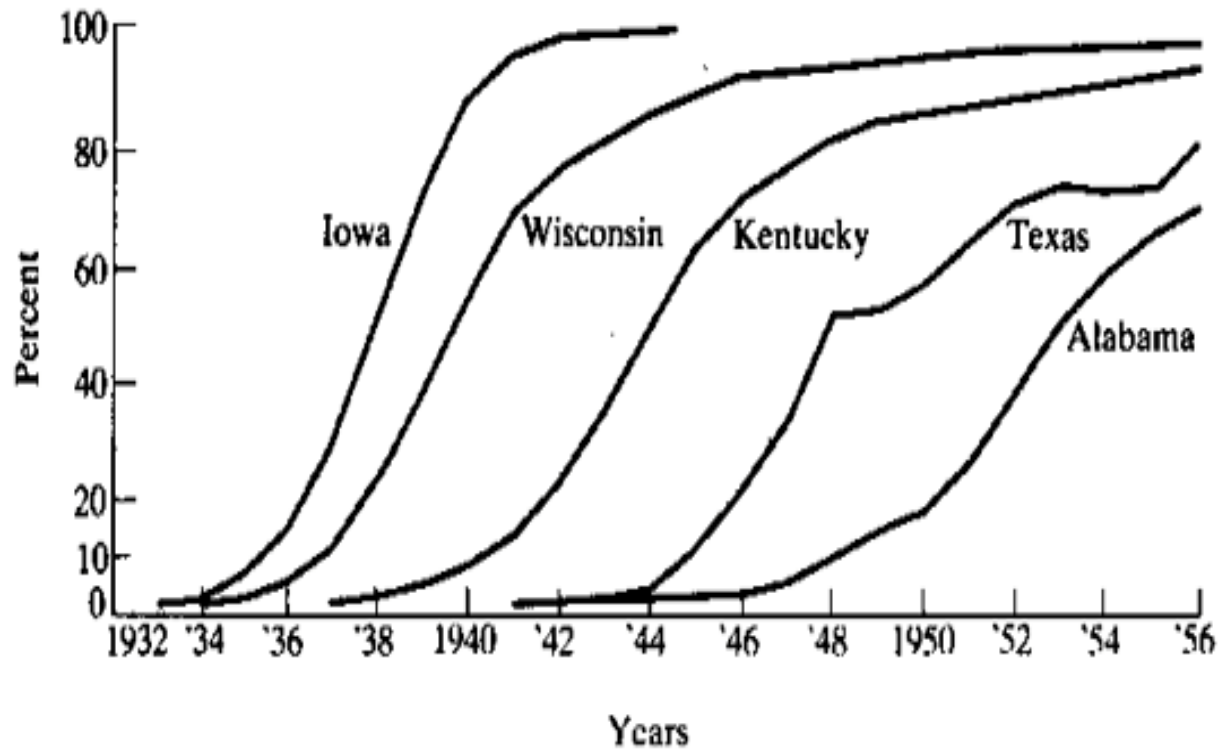


# Diffusion in networks



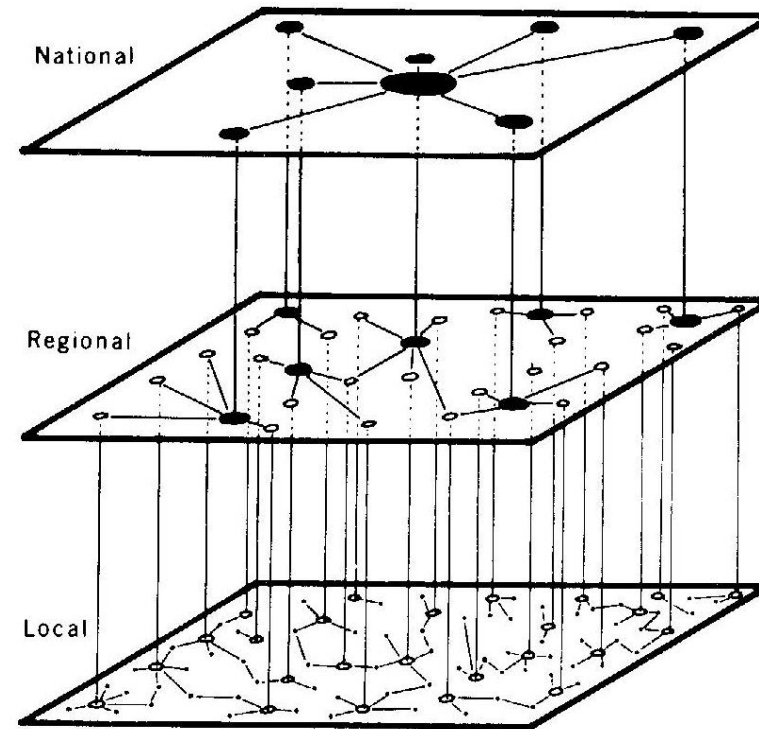
# Spatial diffusion

Innovation gets quicker to proximate places.



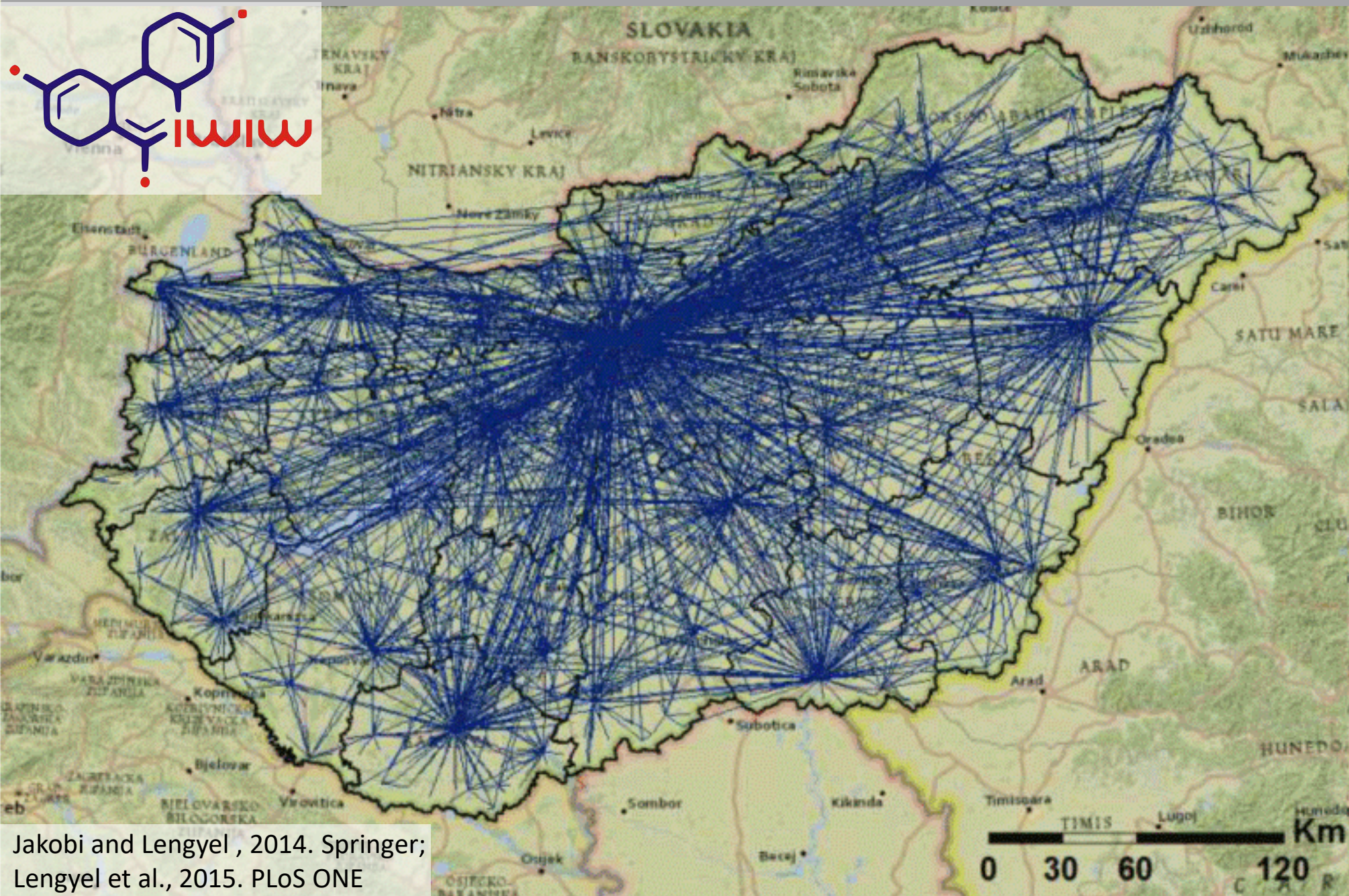
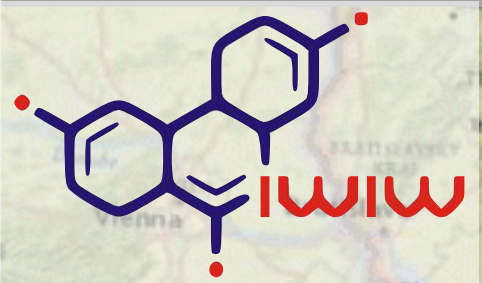
Griliches (1957, Econometrica)

Innovation spreads from big to medium cities and then to small towns.



Haegerstrand (1953, Chicago UP)





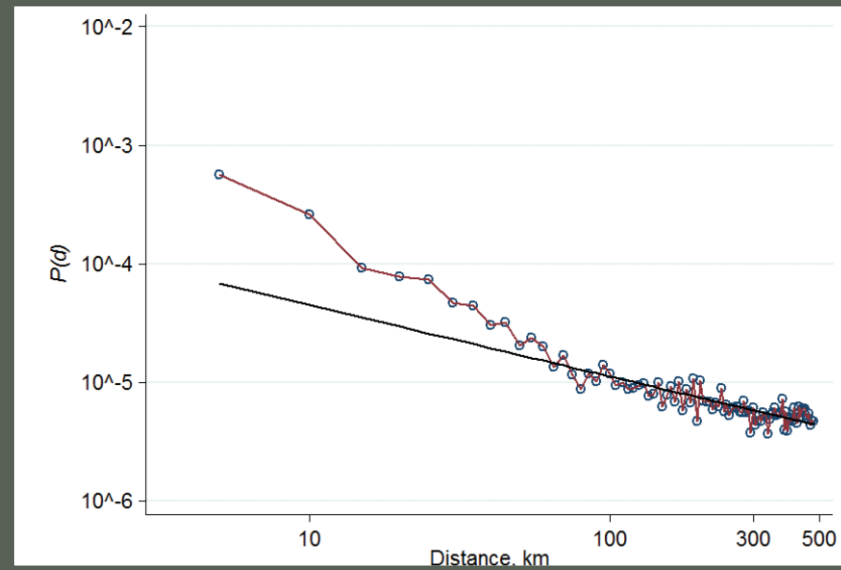
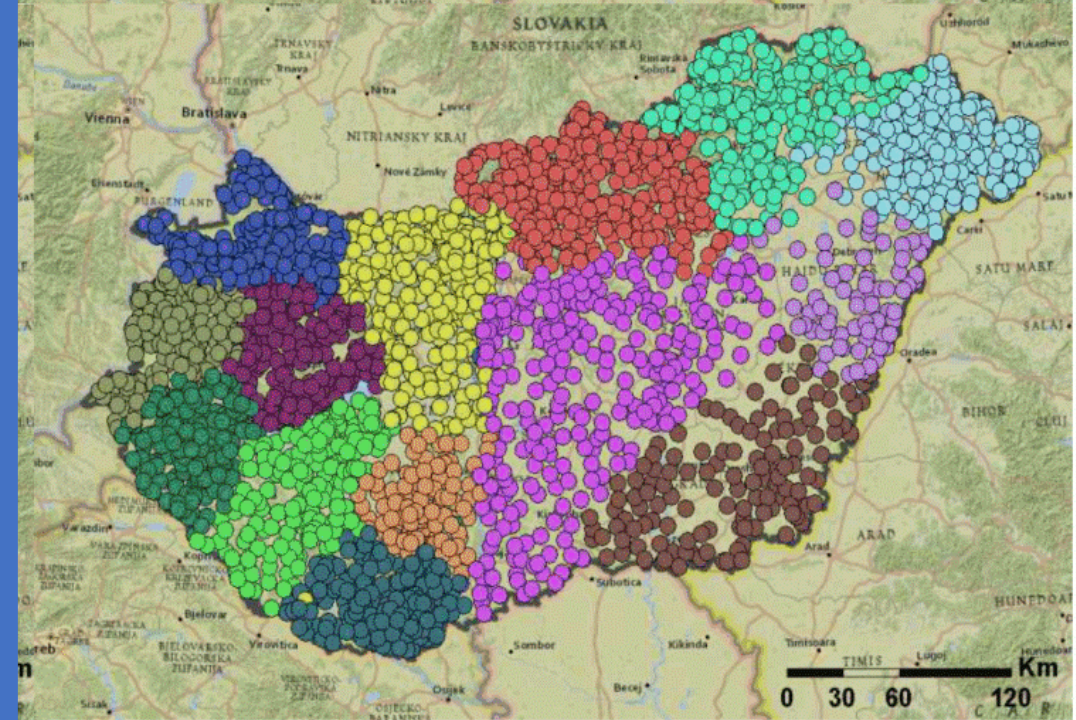
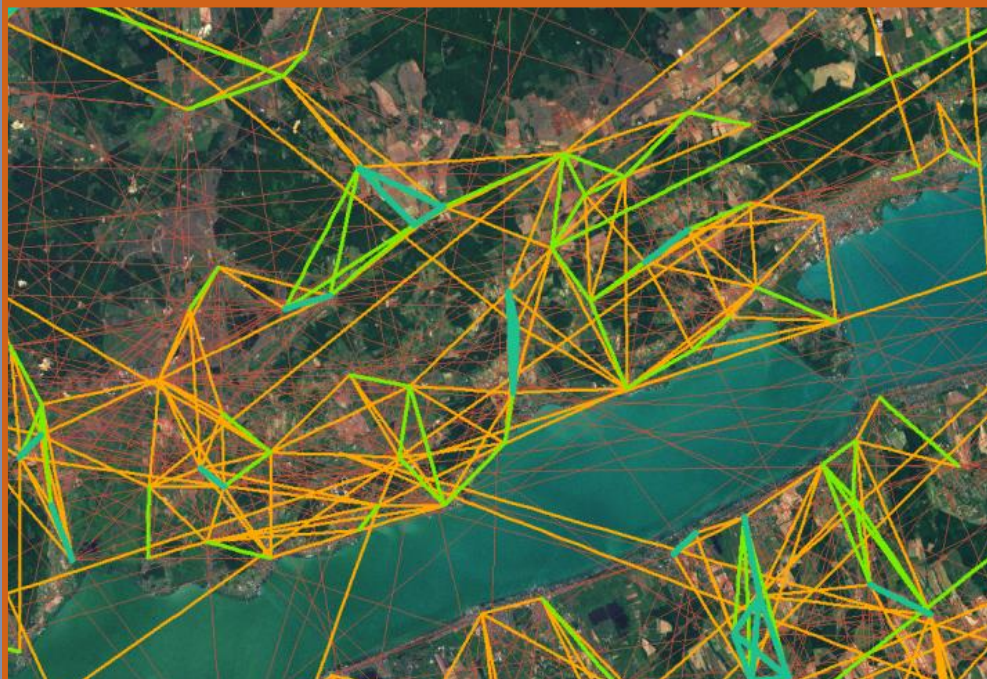
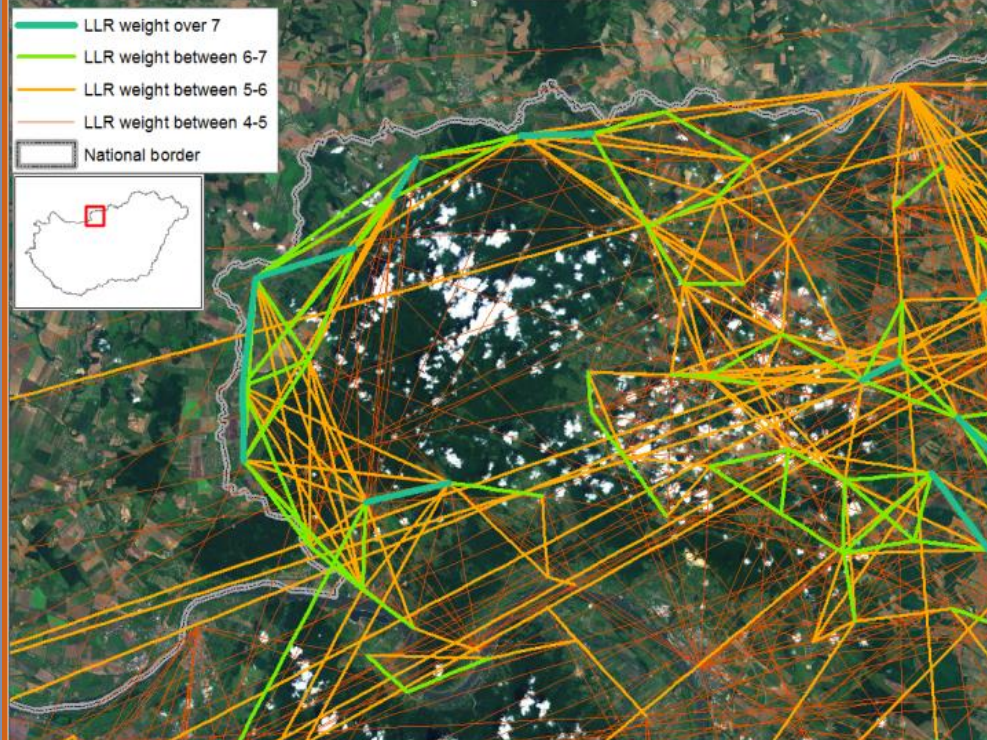
Full life cycle  
of a social  
media  
platform

## Users

- **Location** (self-reported),
- **Date of registration,**
- **Date of last login,**
- **ID of friends,**
- **ID of invitor**

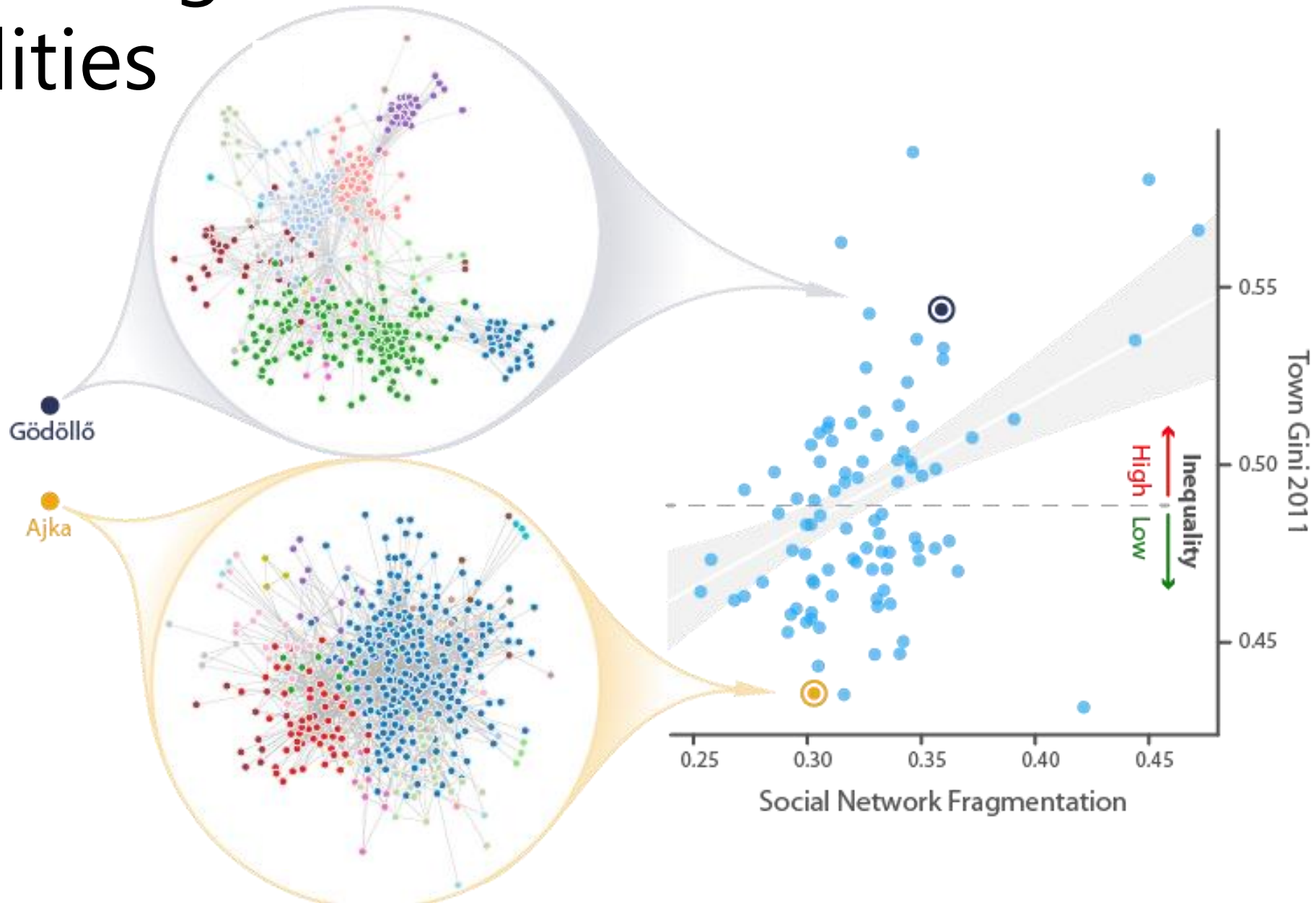


# Geographies of the network





# Network fragmentation is correlated with inequalities

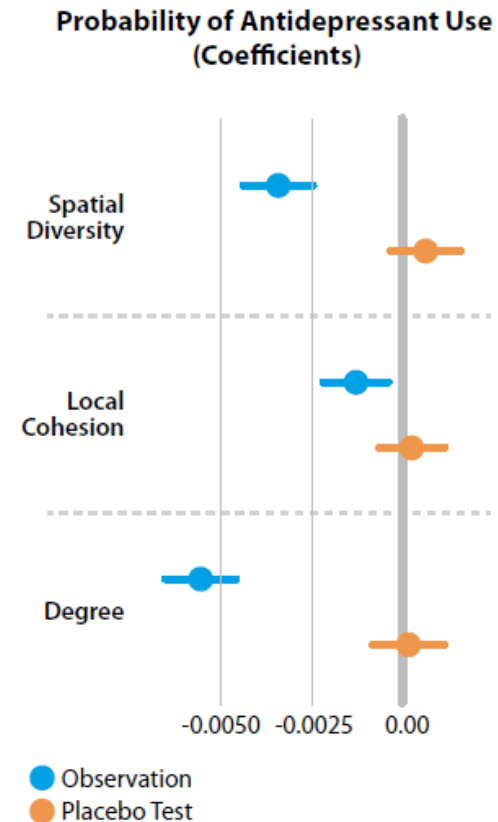
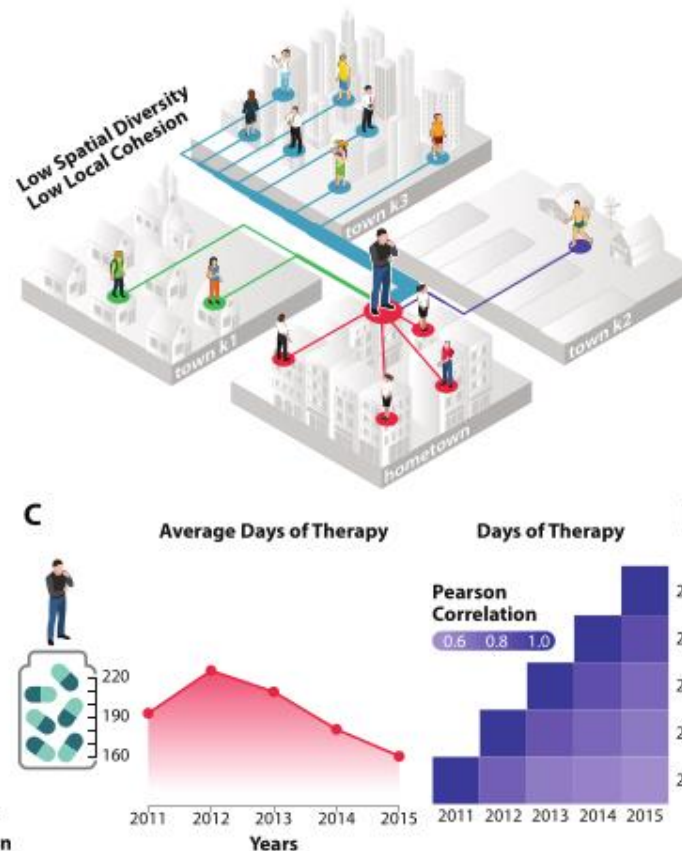
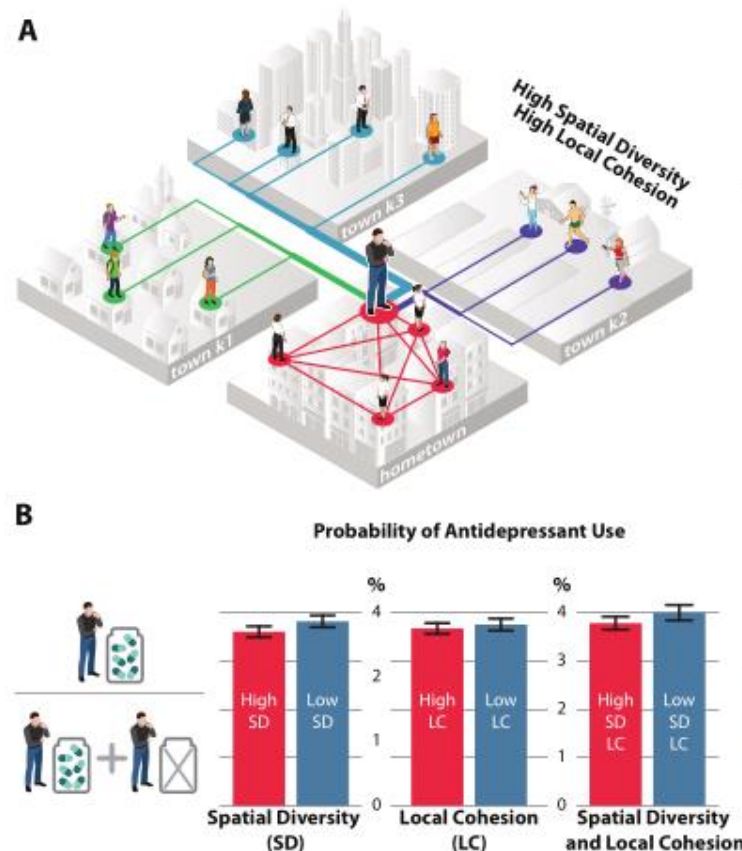
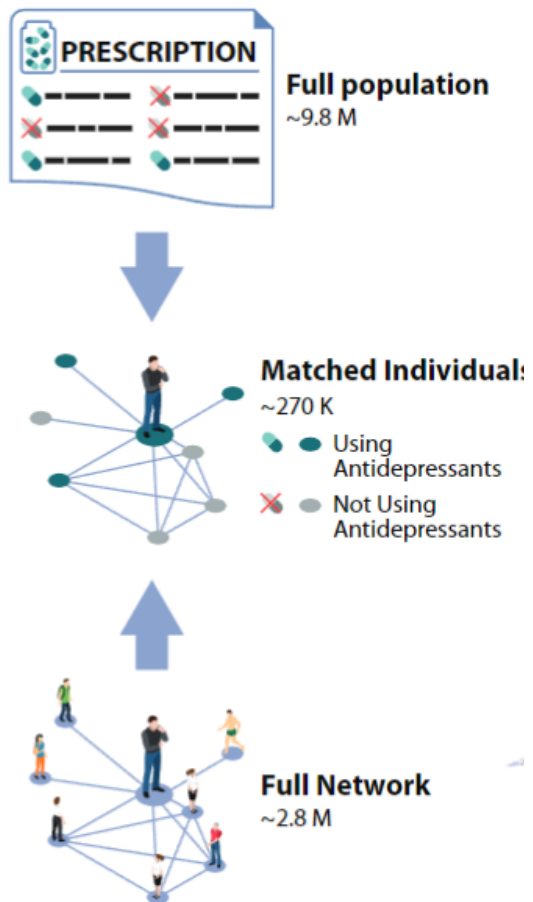


# Diverse spatial networks help mental health

We match the iWiW network with antidepressant purchase.

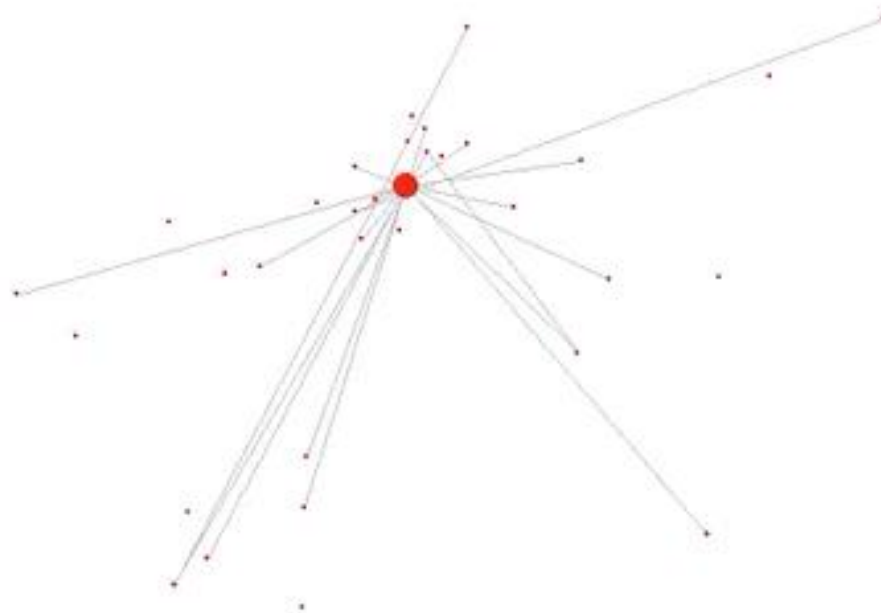
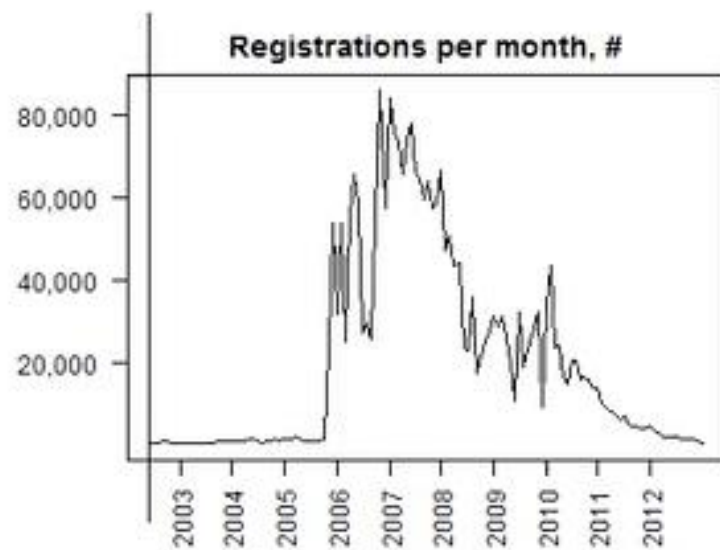
Local Cohesion and Spatial Diversity of social connections capture different qualities of Social Capital.

We use network indicators to estimate antidepressant use.



Registered users, #

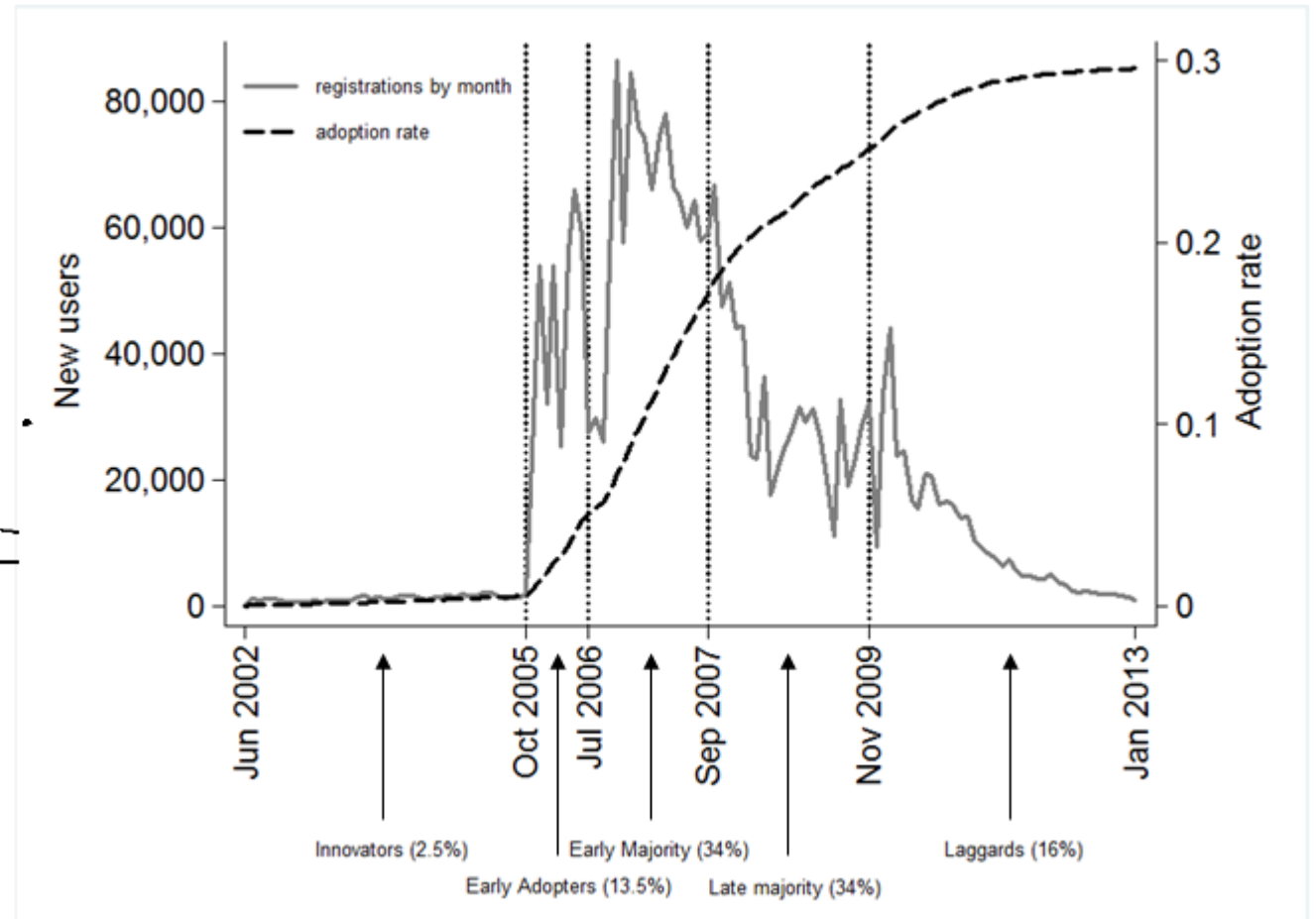
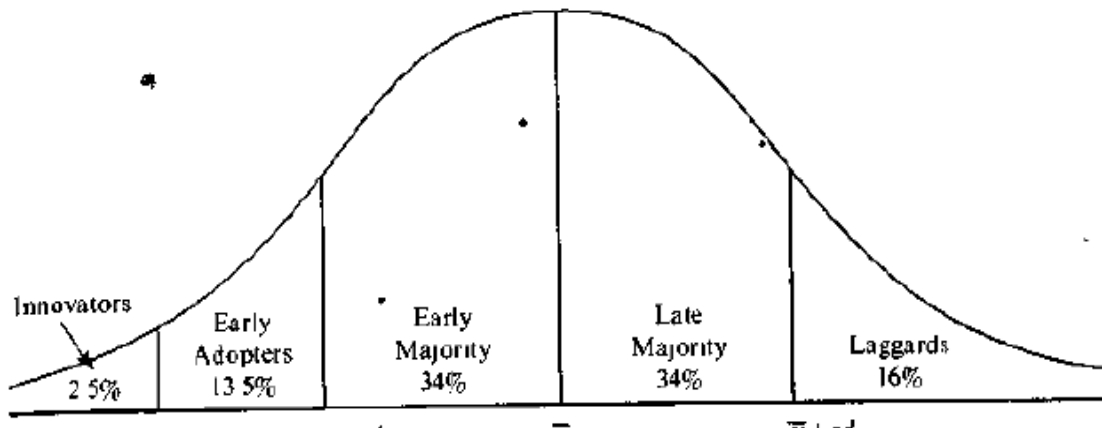
- 10
- 100
- 1000
- 10000
- 100000



Active users, %

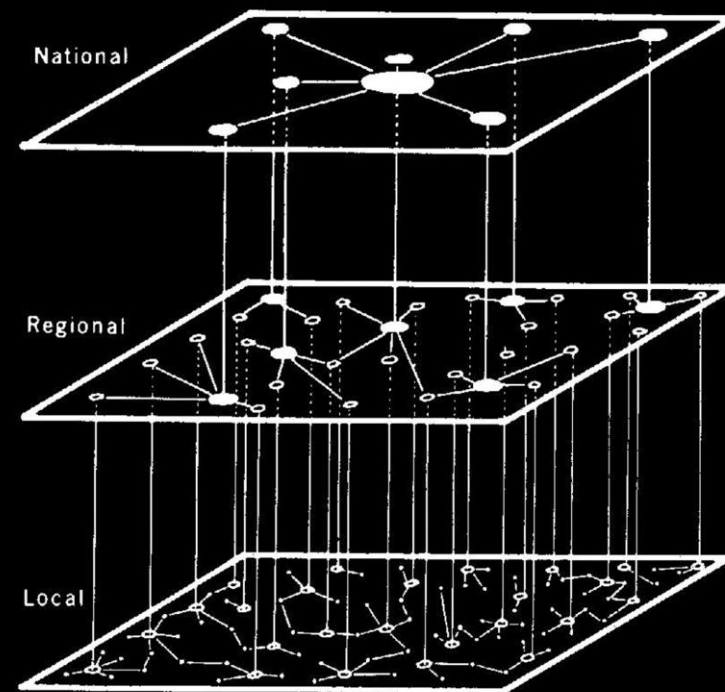
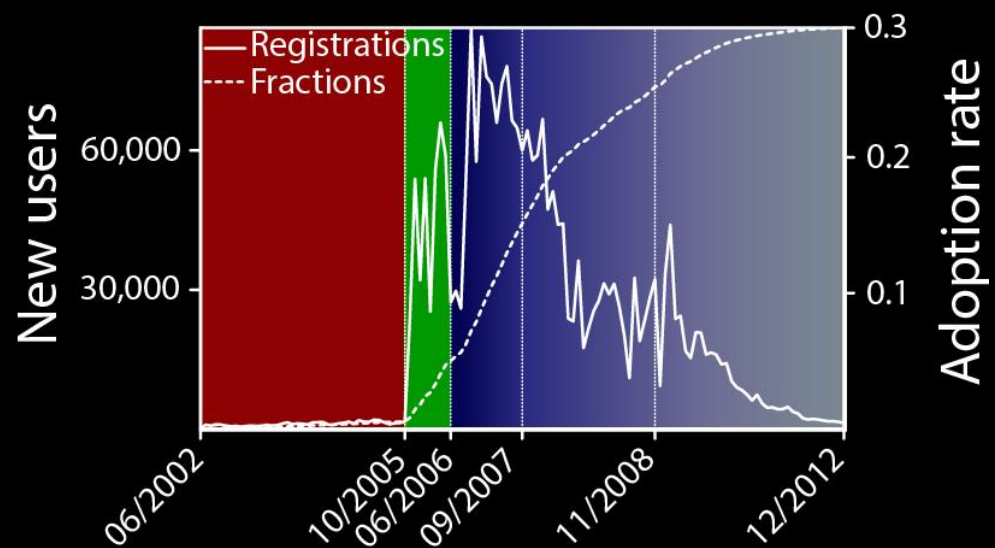


# The iWiW life cycle by Rogers categories



Forrás: Rogers (The Free Press, 1962),

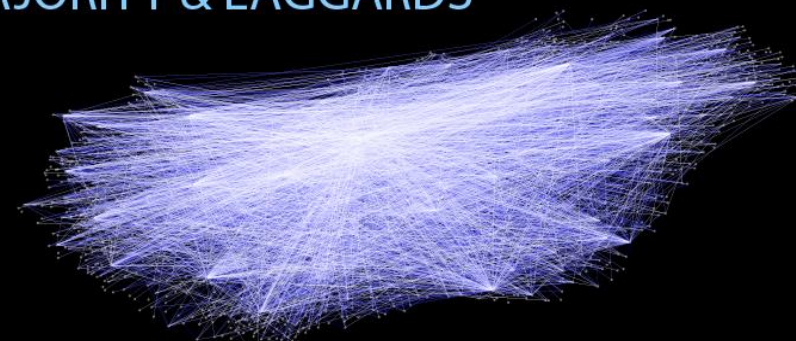
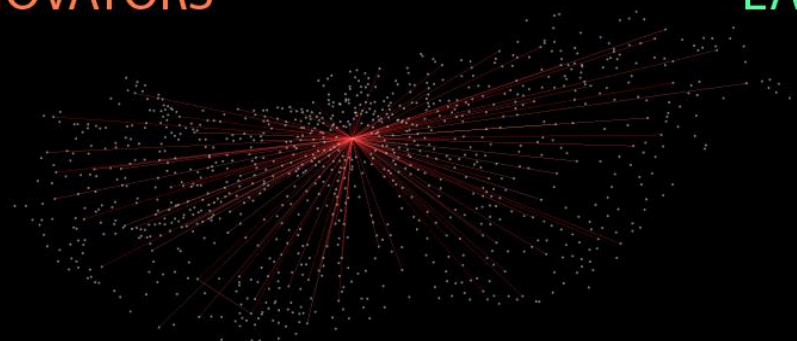




INNOVATORS

EARLY ADOPTERS

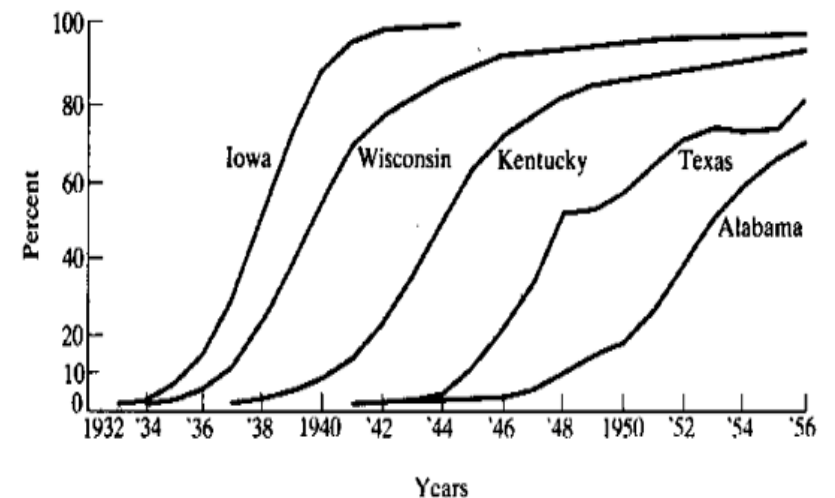
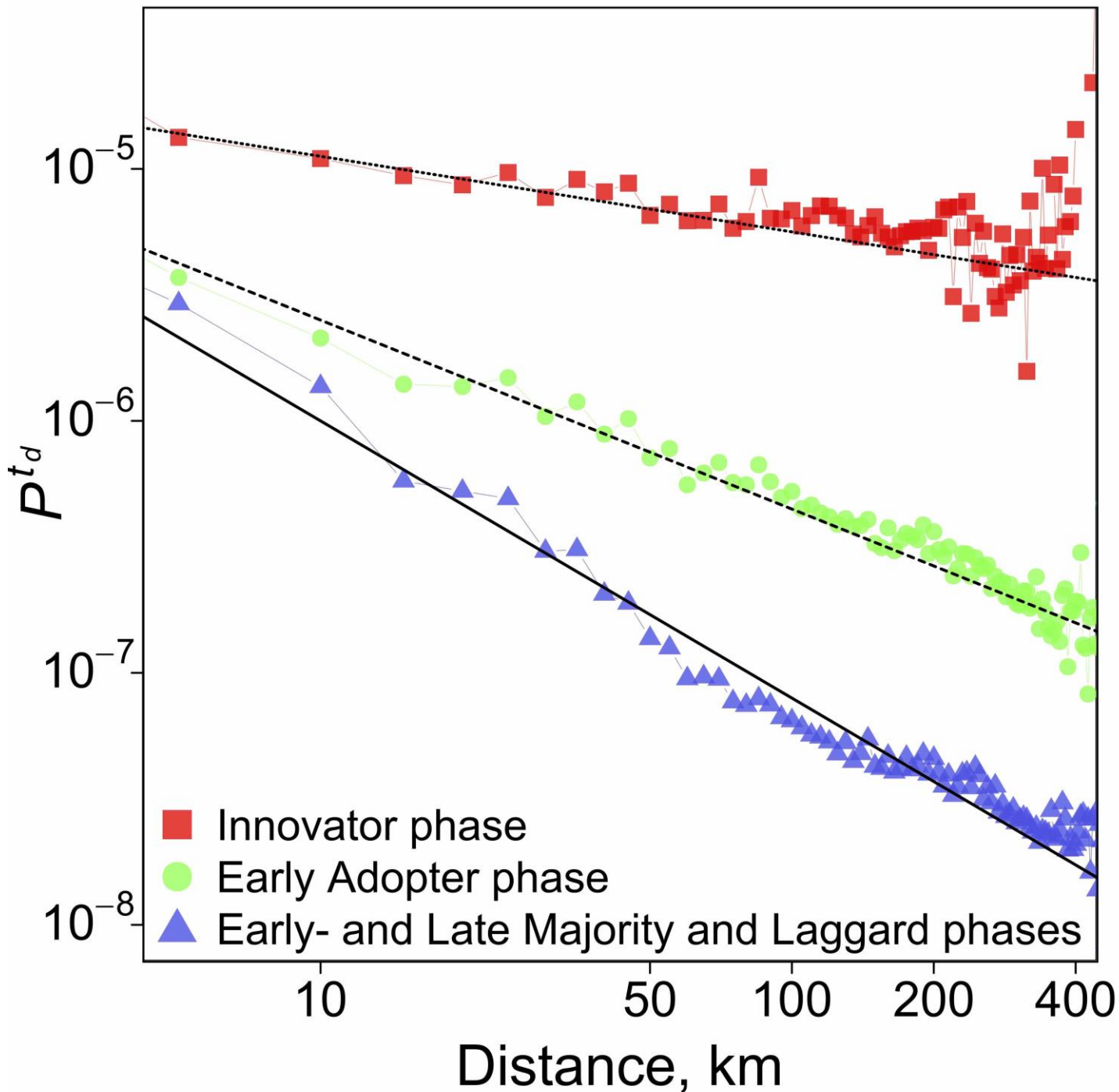
MAJORITY & LAGGARDS

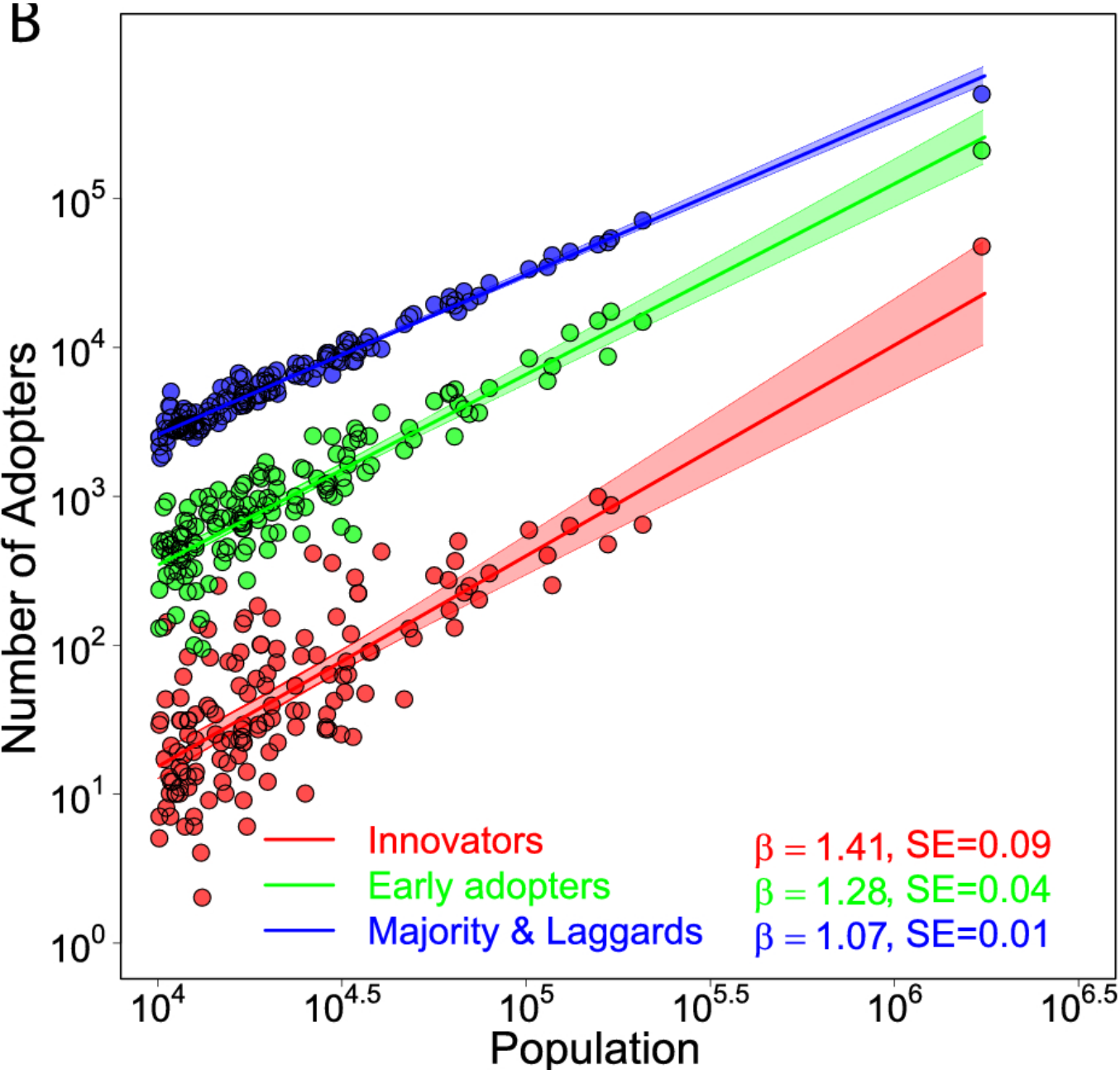


## The role of distance

Distance only slightly decreases the probability of invitation in early phases

Diffusion becomes local in later phases.

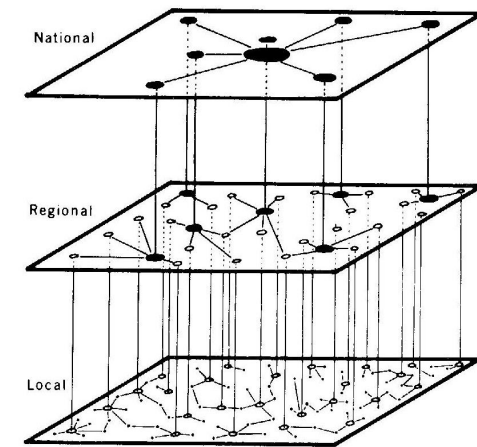




## The role of size

Innovators and Early Adopters concentrate in cities.

Majority and Laggards are proportional to town size.



# Adoption Rate

(Bass Curve)

$$\frac{dy_a(t)}{dt} = (p_a + q_a y_a(t))(1 - y_a(t))$$

$$y_a(t) = m \frac{1 - e^{-(p_a + q_a)t}}{1 + \frac{q_a}{p_a} e^{-(p_a + q_a)t}}$$

# of adopters

Innovation Par.

$y_a(t)$

$p_a$

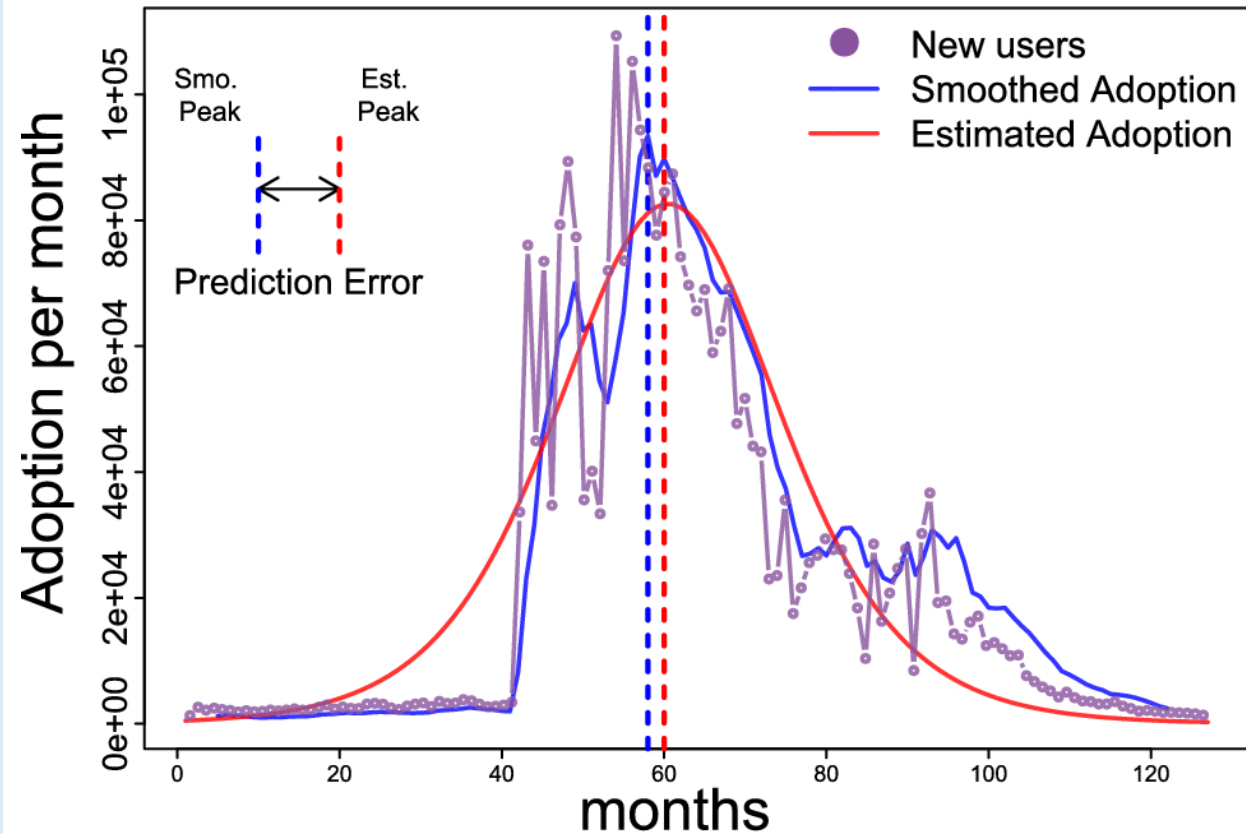
$t$

$q_a$

Time (months)

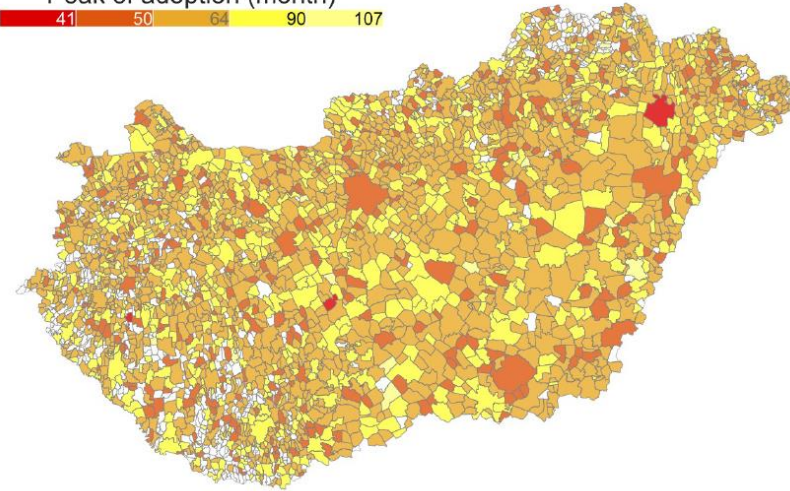
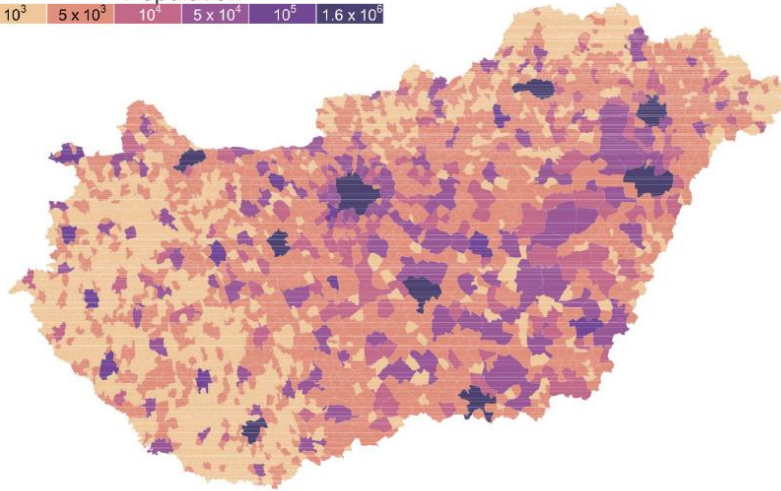
Imitation Par.

A

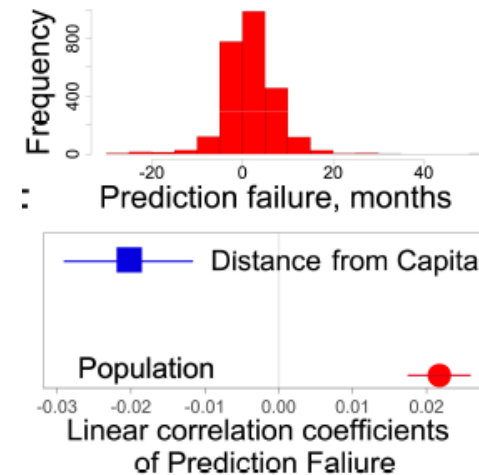
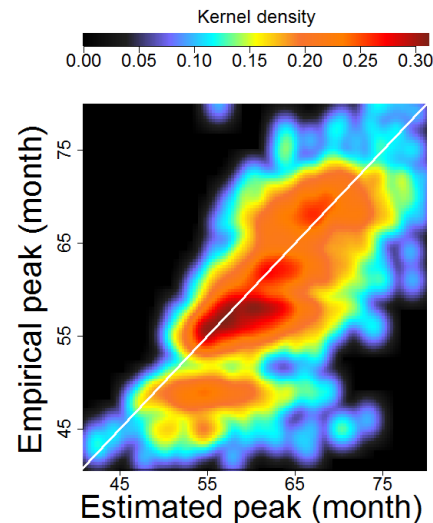




# Local adoption peaks and prediction errors

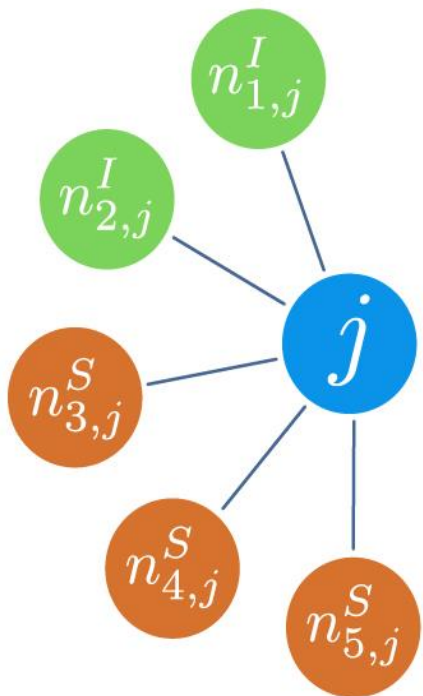


$$T_{peak-t}^i = \frac{\ln p_a^i + \ln q_a^i}{p_a^i + q_a^i}$$

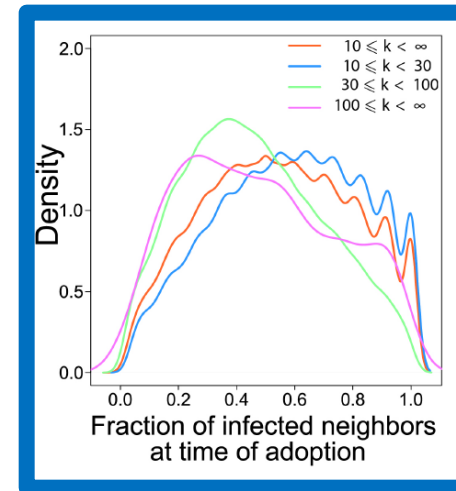


# Agent-based model of diffusion in the network

- We fix the network.
- Inactives are not removed.
- 10% sample (300 K users)
- Sampling stratified by towns and network moduls



Innovation or  
Marketing  
parameter



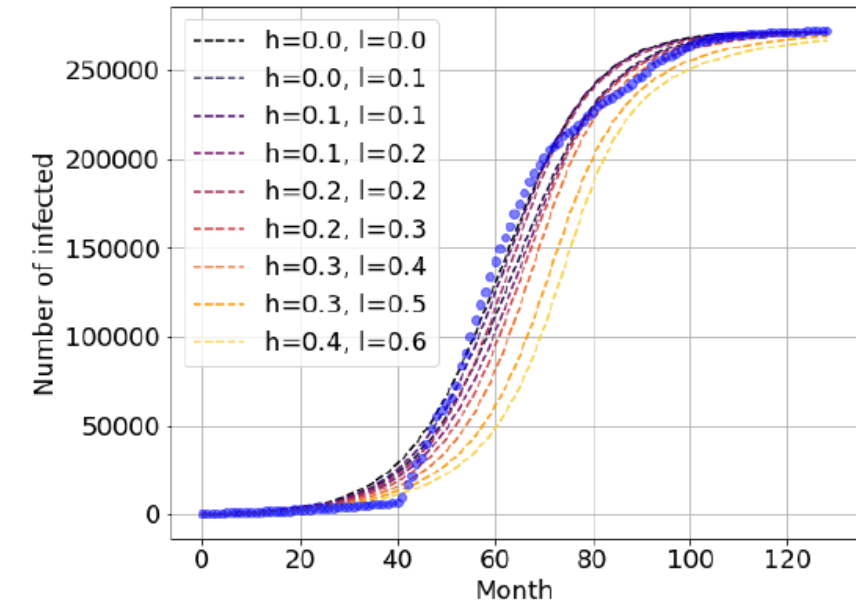
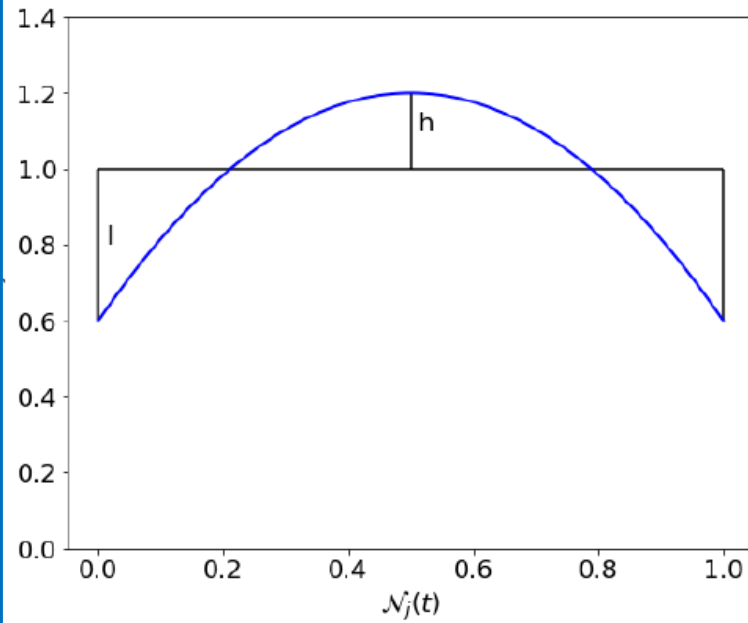
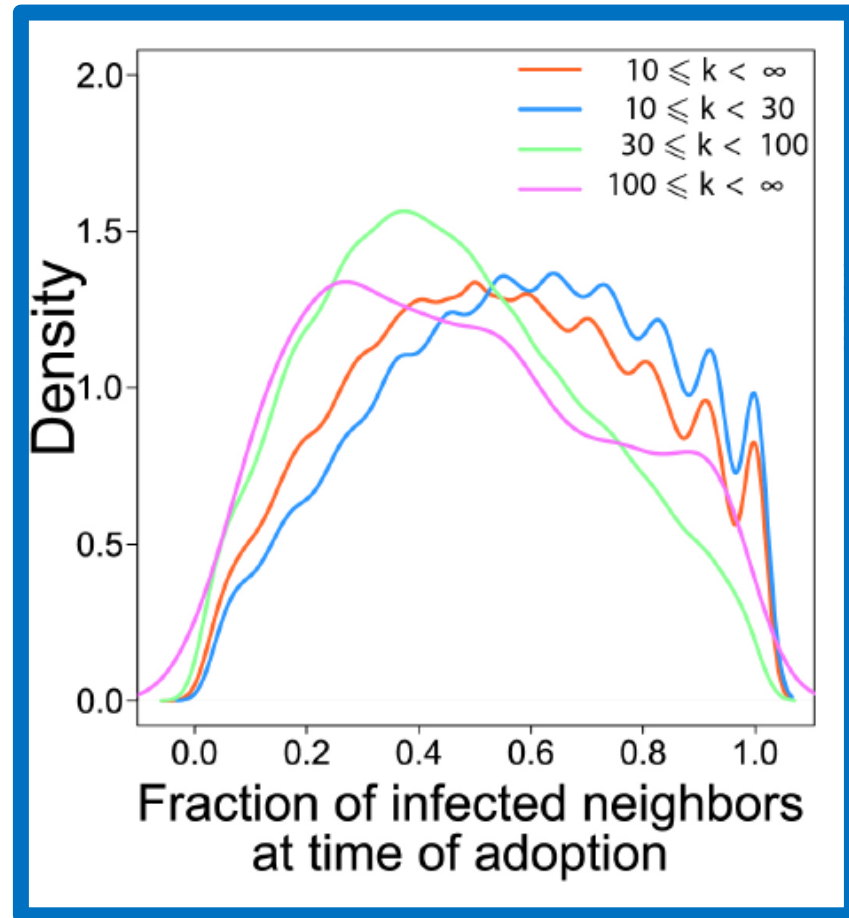
Network effect

$$\mathcal{N}_j(t) = \frac{\#n_j^I(t)}{\#n_j^I(t) + \#n_j^S(t)}$$

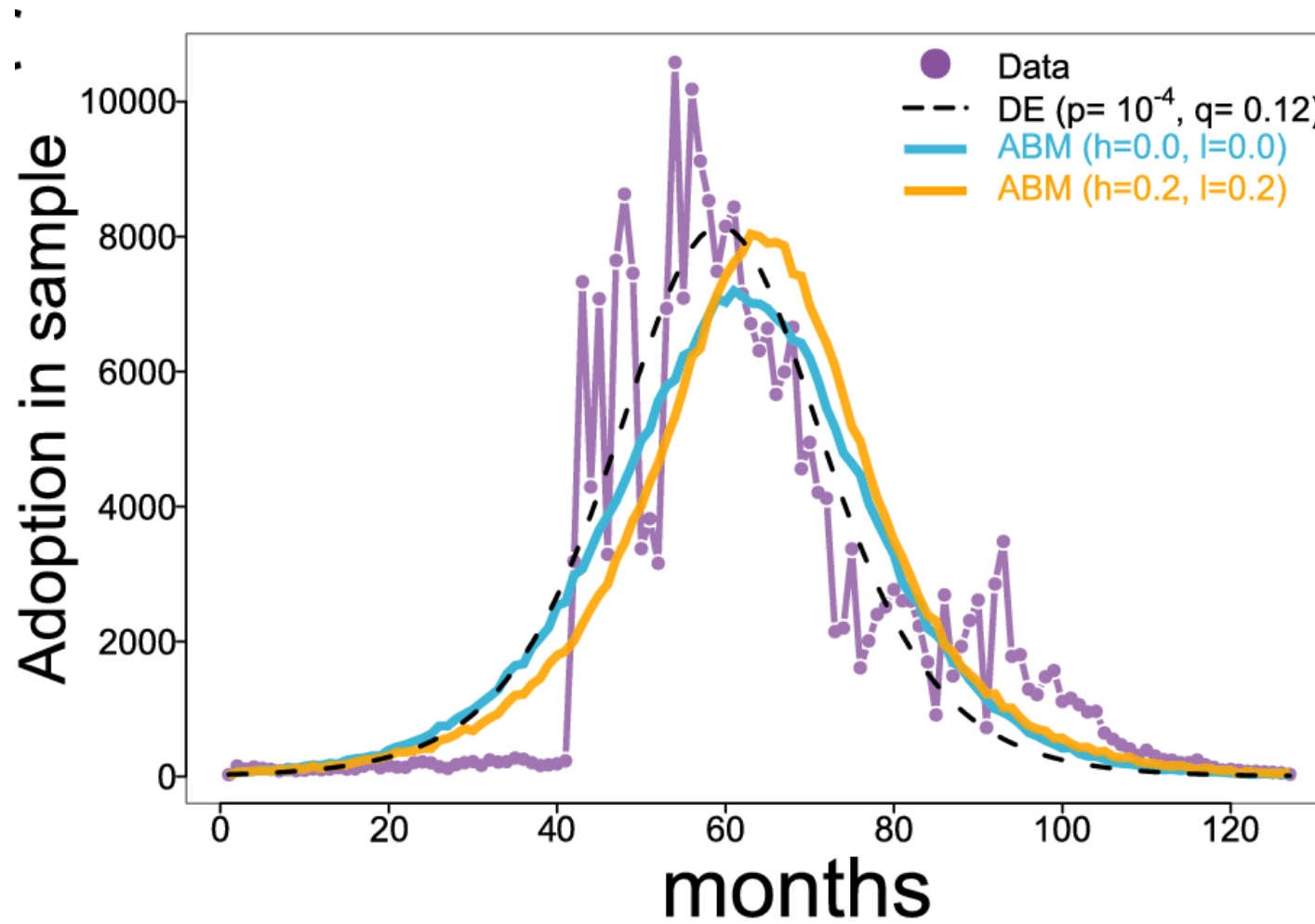
$$F_j(t+1) = \begin{cases} I & \text{if } U(0,1)_{jt} < \hat{p}^{\text{ABM}} + T(\mathcal{N}_j(t), h, l) \times \mathcal{N}_j(t) \times \hat{q}^{\text{ABM}} \\ S & \text{otherwise} \end{cases}$$

Imitation parameter

# Threshold parameters

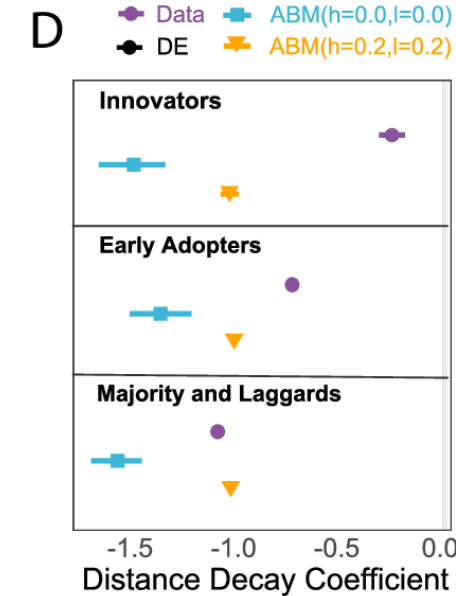
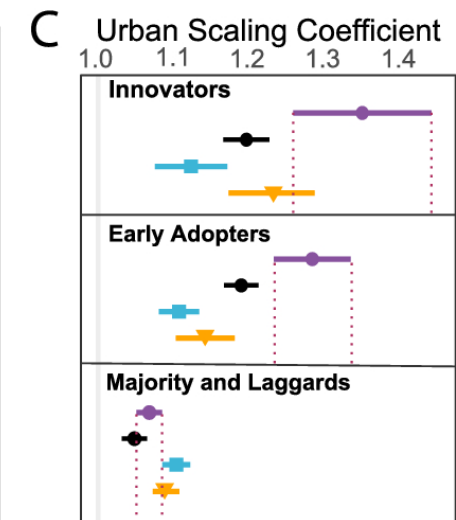
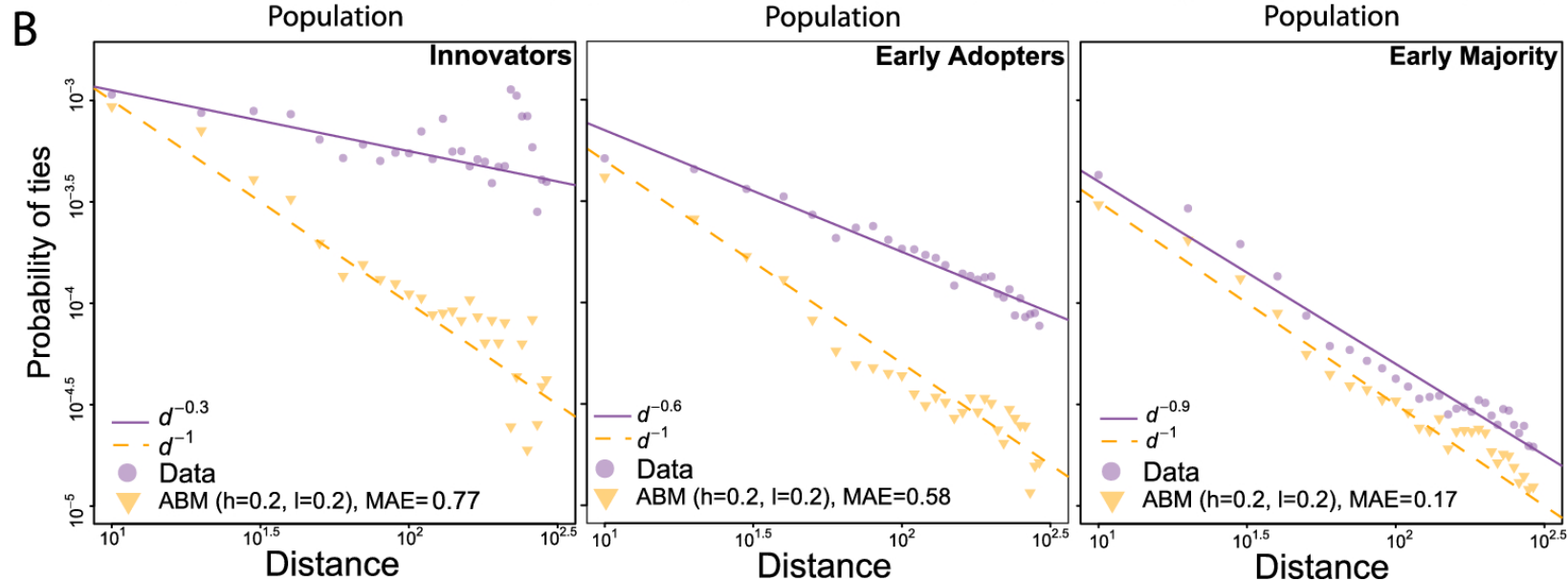
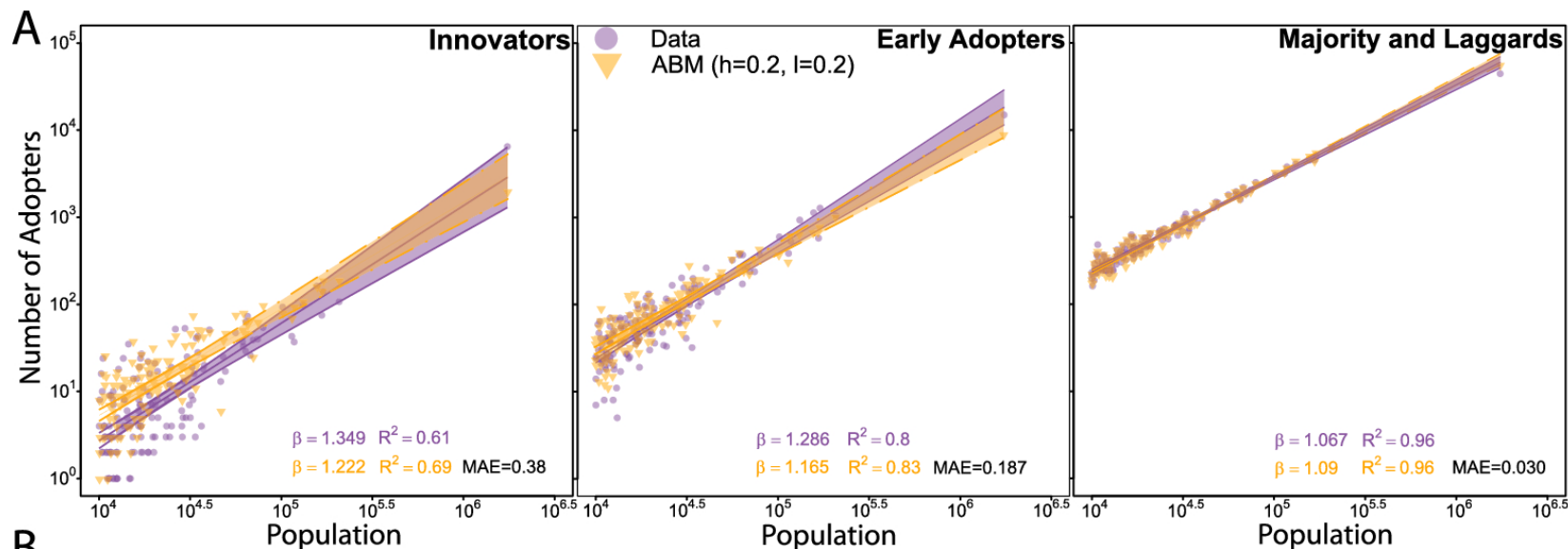


# Results

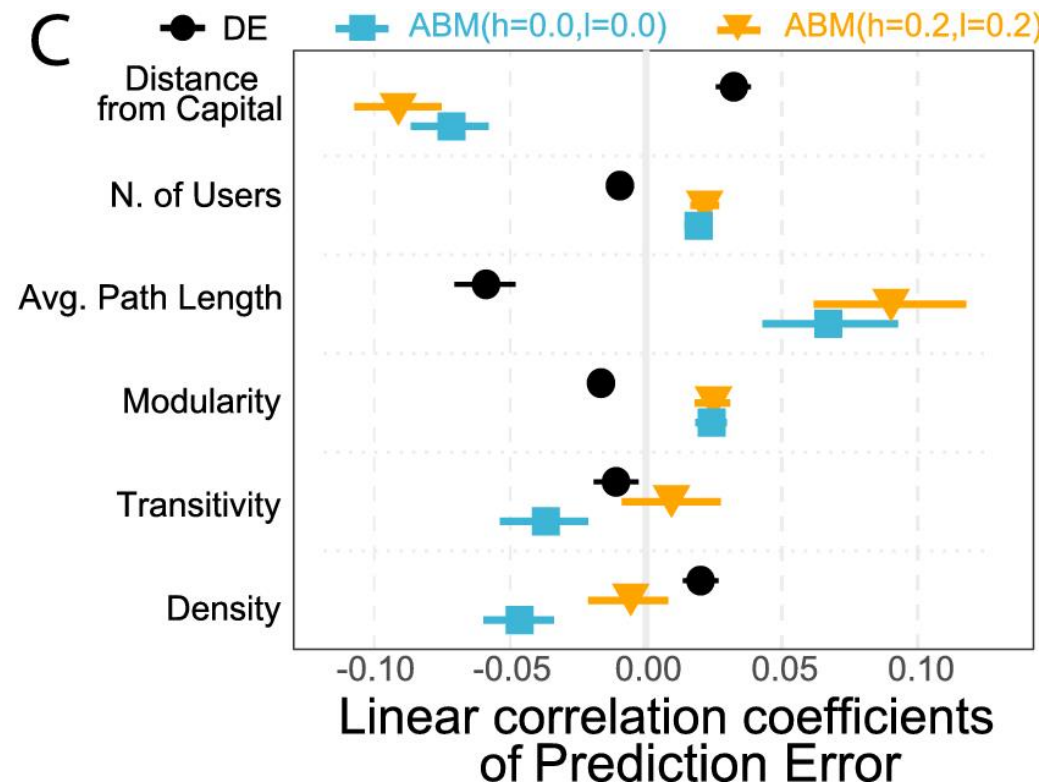
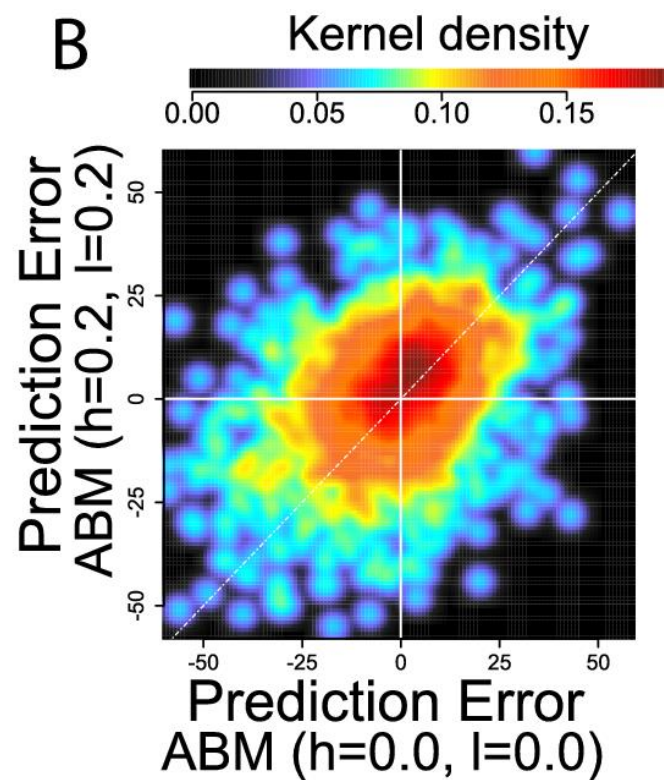
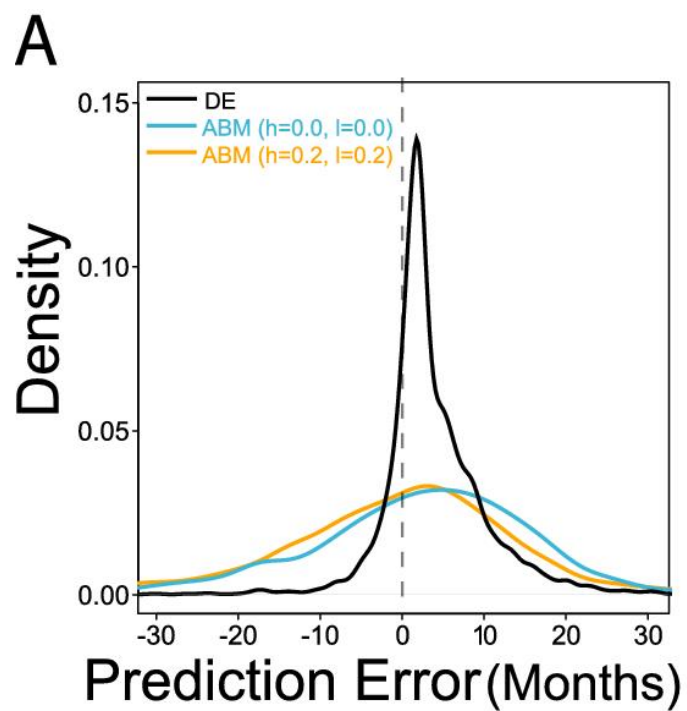




# Results



# Results



# The spatial life cycle hypotheses

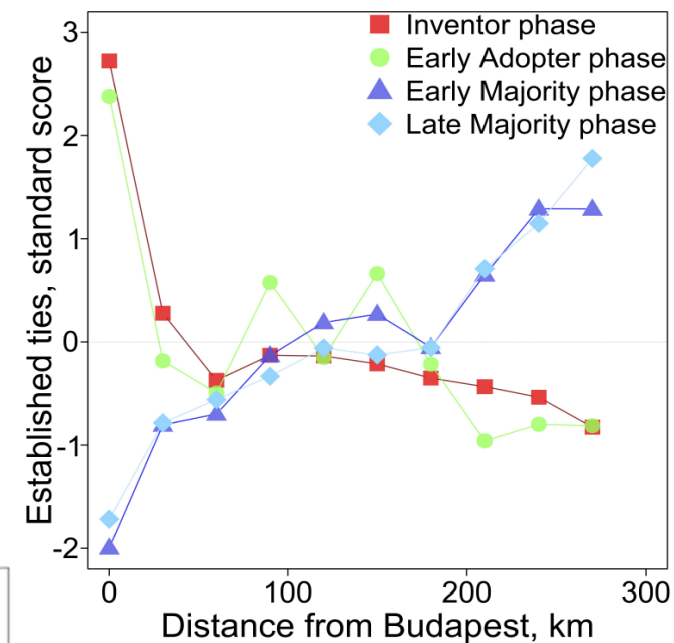
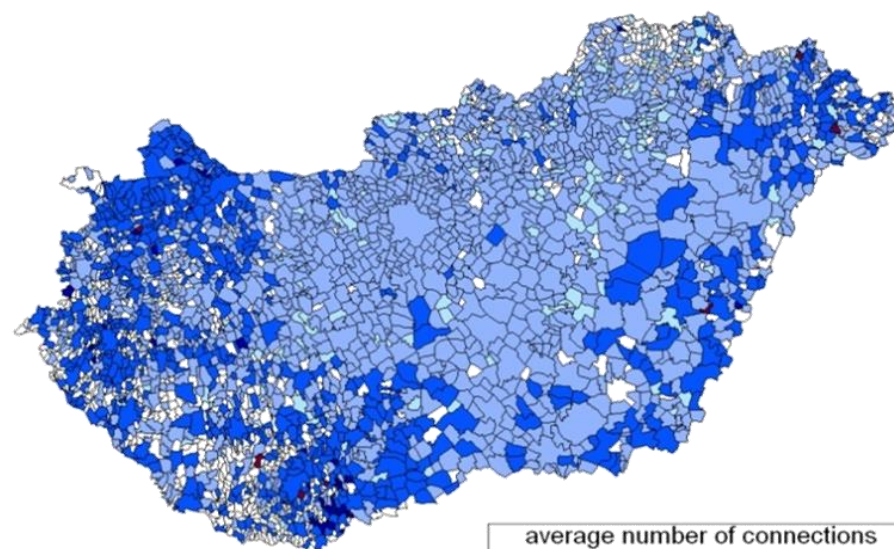
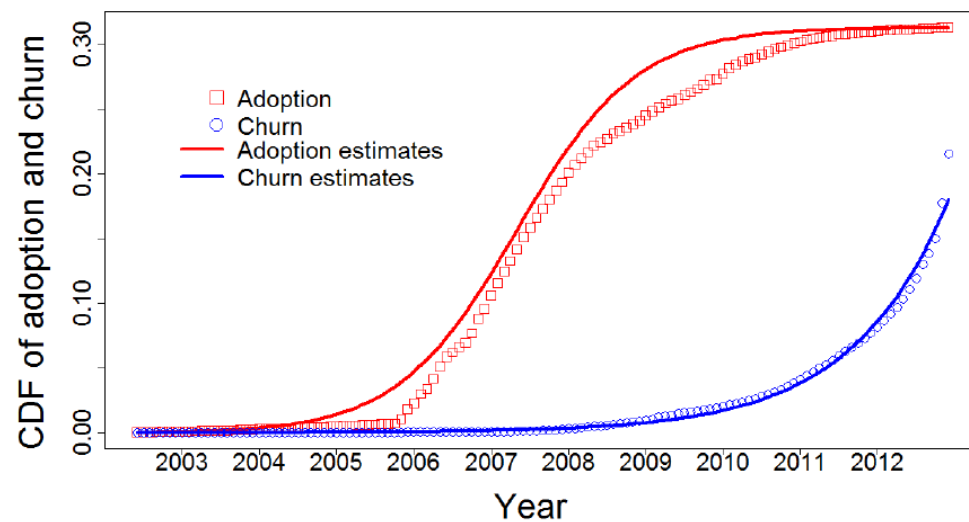
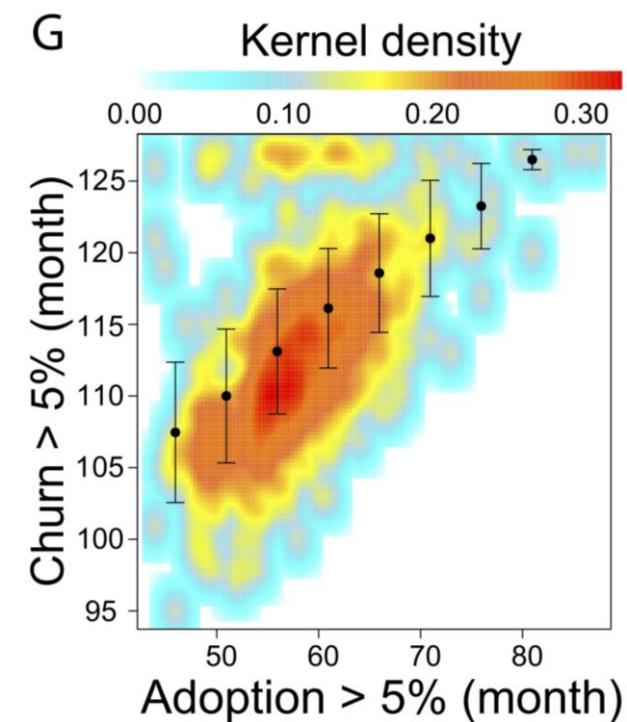
Balázs Lengyel

ANETI Lab ELTE CERS and Corvinus Institute for Advanced Studies  
Institute for Data Analytics and Information Systems, Corvinus University





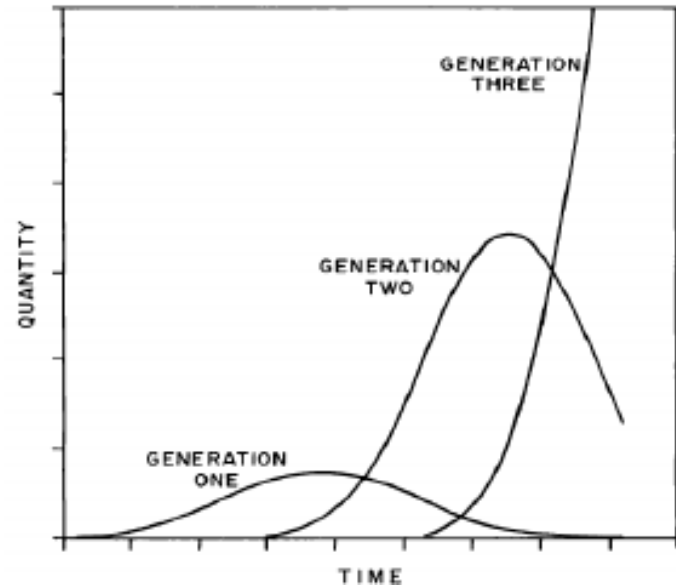
# CHURN





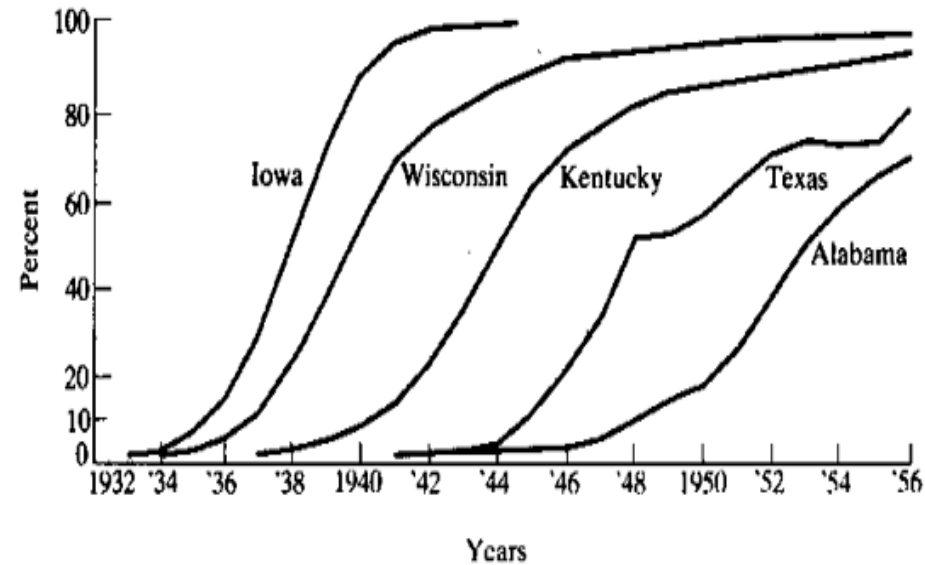
# Waves of technology life-cycle

New technologies outcompete old ones.



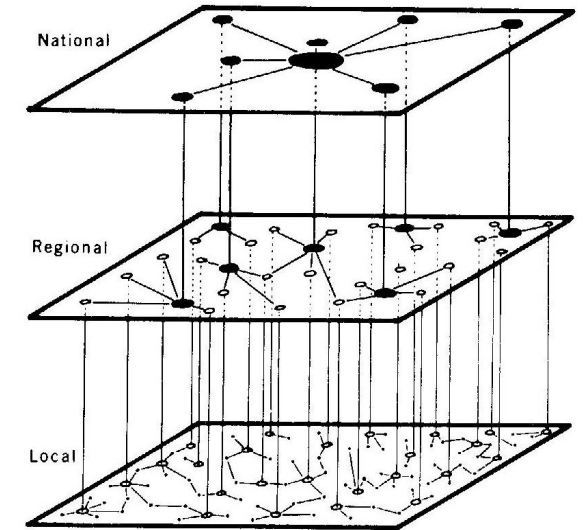
Norton & Bass (1987, Man Sci)

Innovation gets quicker to proximate places.



Griliches (1957, Econometrica)

Innovation spreads from big to medium cities and then to small towns.



Haegerstrand (1953, Chicago UP)

Spatial wave of  
adoption and  
churn



# The spatial life cycle hypotheses

H1: *„The churn from an innovation is early in places that have adopted it early.“*

- Declining relative advantage and social value after saturation
  - Adoption in periphery happens when centrum saturates
  - Spatial differences in perceived novelty
  - Value fades away
- 
- Assumption: like adoption, churn also happens in identical spatial networks
  - Mixture of diffusion of obsolescence, new waves, local network economies

# The spatial life cycle hypotheses

H2: *„Not only adoption but also churn starts in large cities.”*

- Early adoption and churn are both urban phenomena.
- Cities are more innovative and more susceptible for innovation.
- Saturation is early – obsolescence is also early.
- New products can also arrive early.



# The spatial life cycle hypotheses

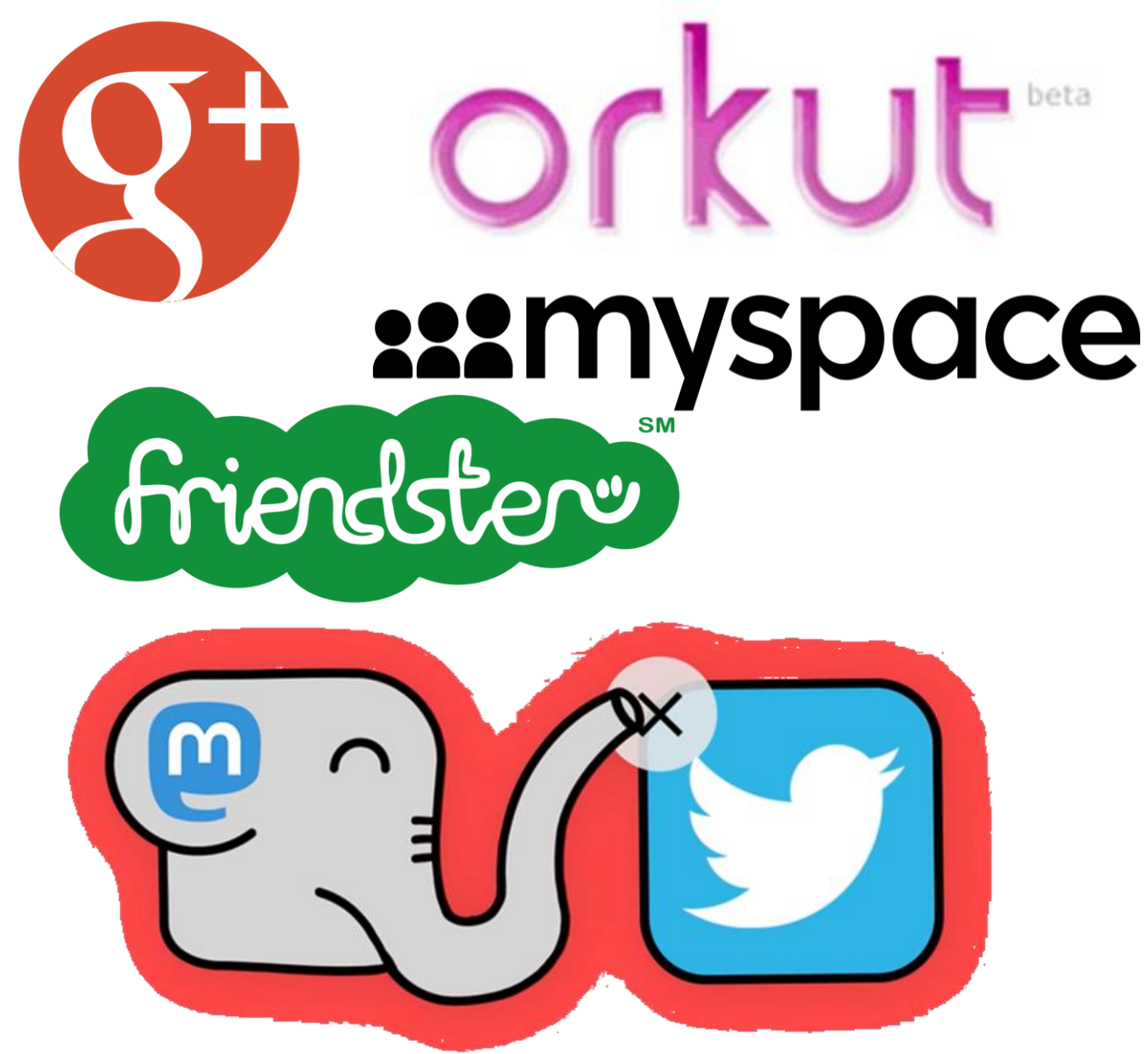
H3: *„Churn is earlier in those locations, where the adoption of the replacement innovation cycle is early.”*

- New technologies are adopted on the basis of the old one.
- Spatial diffusion is similar across waves.
- The new replaces the old.

# The spatial life cycle hypotheses

H4: *„The innovation’s embeddedness in local socio-technological systems prolong its life cycle in places.”*

- Local network economies slow down churn.
- Benefits of the old technology prevail if it is embedded in local systems.
- Diffusion and churn happens in dynamic networks: local interdependencies emerge
  - Early adopters have more time to develop interdependencies
  - The networks of late adopters function more as channels of obsolescence diffusion



## Not just social media:

(H2) Innovation is concentrated in cities  
(Bettencourt et al. 2007)

(H3) All IT waves come from the Bay area.

(H4) Tightly connected innovation systems tend to stick to old technologies (Grabher, 1993).

This path dependency is strong in networked products.

|                                               |                 |                            |
|-----------------------------------------------|-----------------|----------------------------|
| Jane Doe<br>123 Main St<br>Anywhere US 10111  |                 | Date <u>07/01/2018</u> 790 |
| PAY TO THE ORDER OF <u>ACME Grocery Shop</u>  |                 | \$ <u>8.15</u>             |
| <u>EIGHT AND 15/100</u> DOLLARS               |                 |                            |
| Your Bank<br>456 Main St<br>Anywhere US 10111 |                 |                            |
| MEMO <u>Lunch w/Friends</u>                   | <u>Jane Doe</u> |                            |
| ① ② ③ ④ ⑤ ⑥                                   |                 |                            |

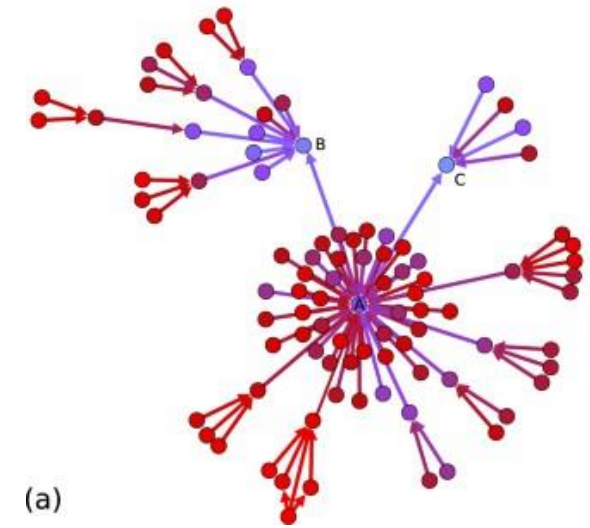
# Previous research on social influence in churn

Lőrincz et al. 2019 Social Networks:

- Central individuals churn later – the network is of greater value for them.

Török and Kertész 2017, SRep

- The cascade of churn starts from the peripheral actors and their churn triggers central actors.



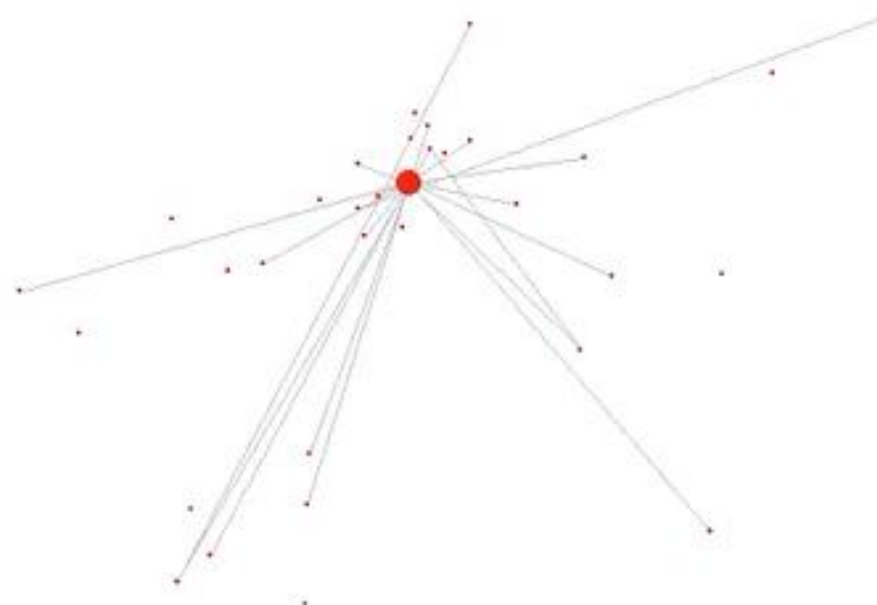
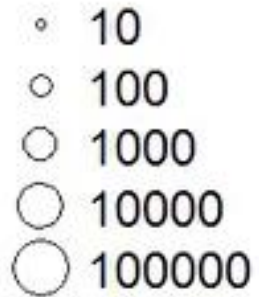
(a)

Török and Kertész (2017, Scientific Reports)



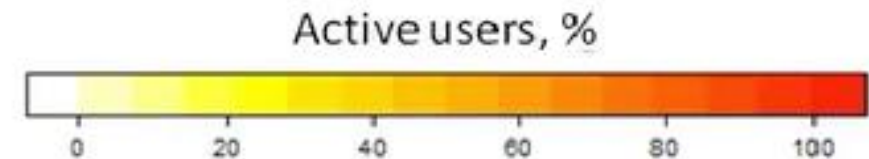
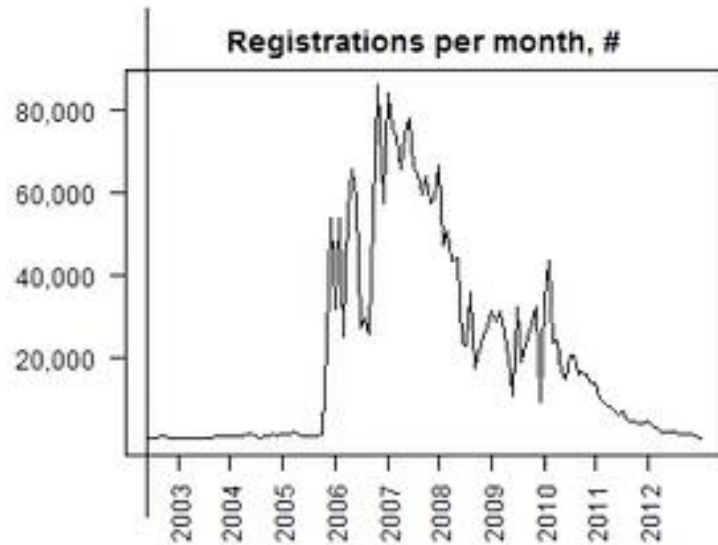
# Full life-cycle of a social media platform

Registered users, #

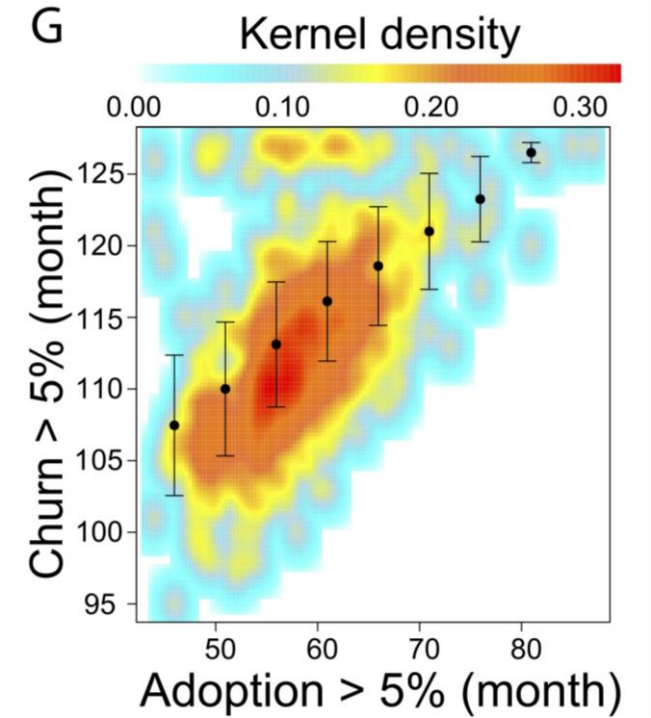
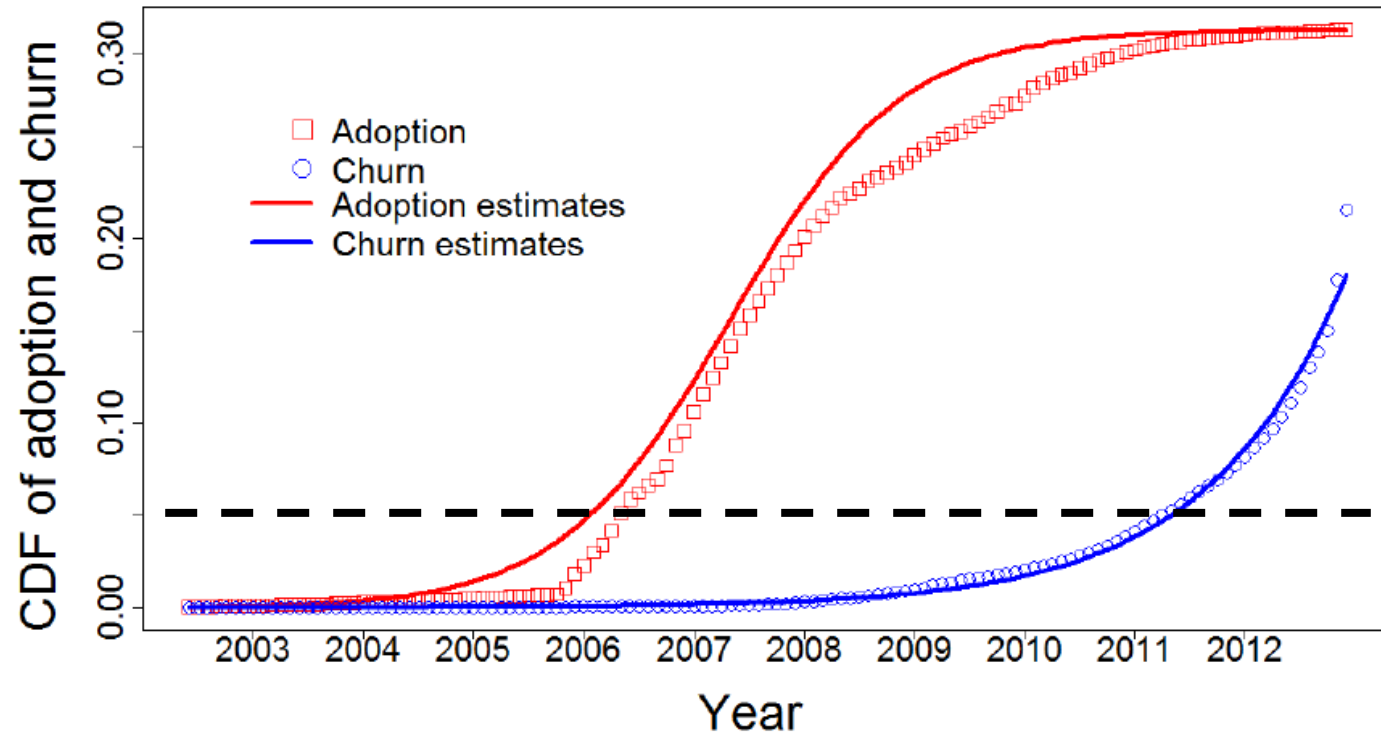


2.8 M Users

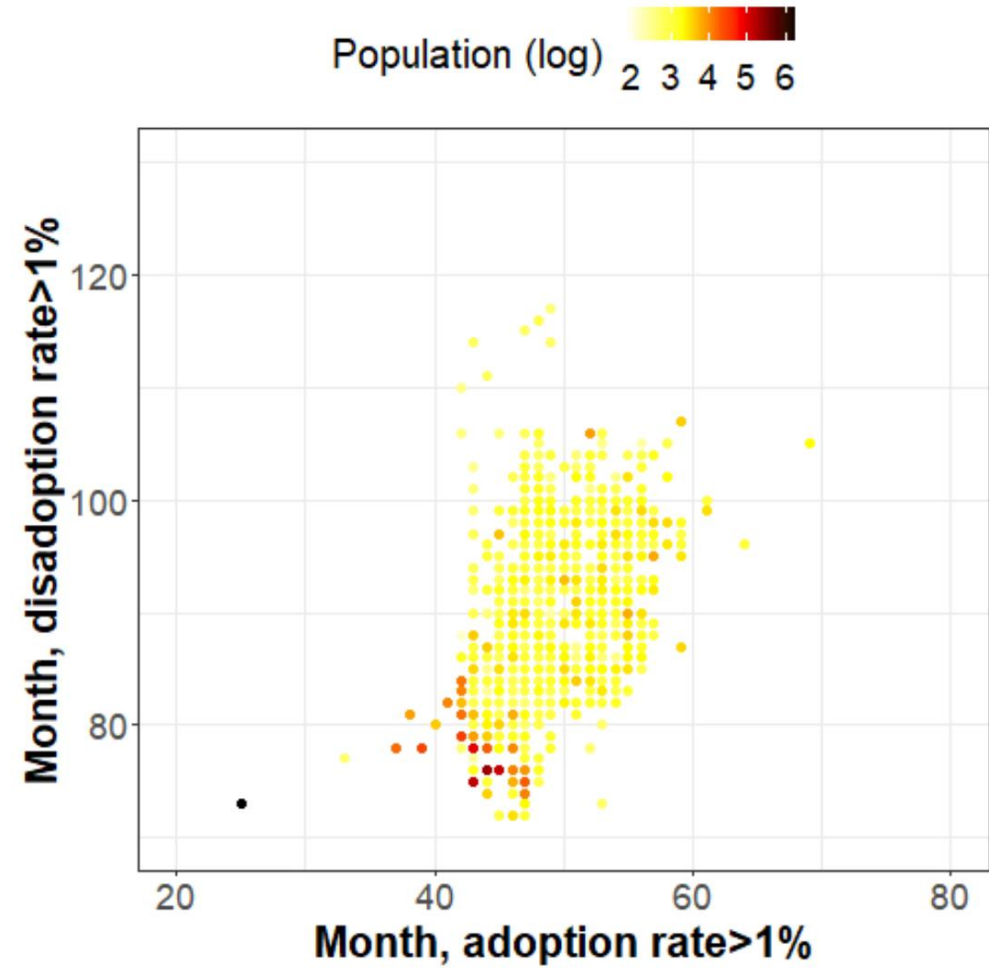
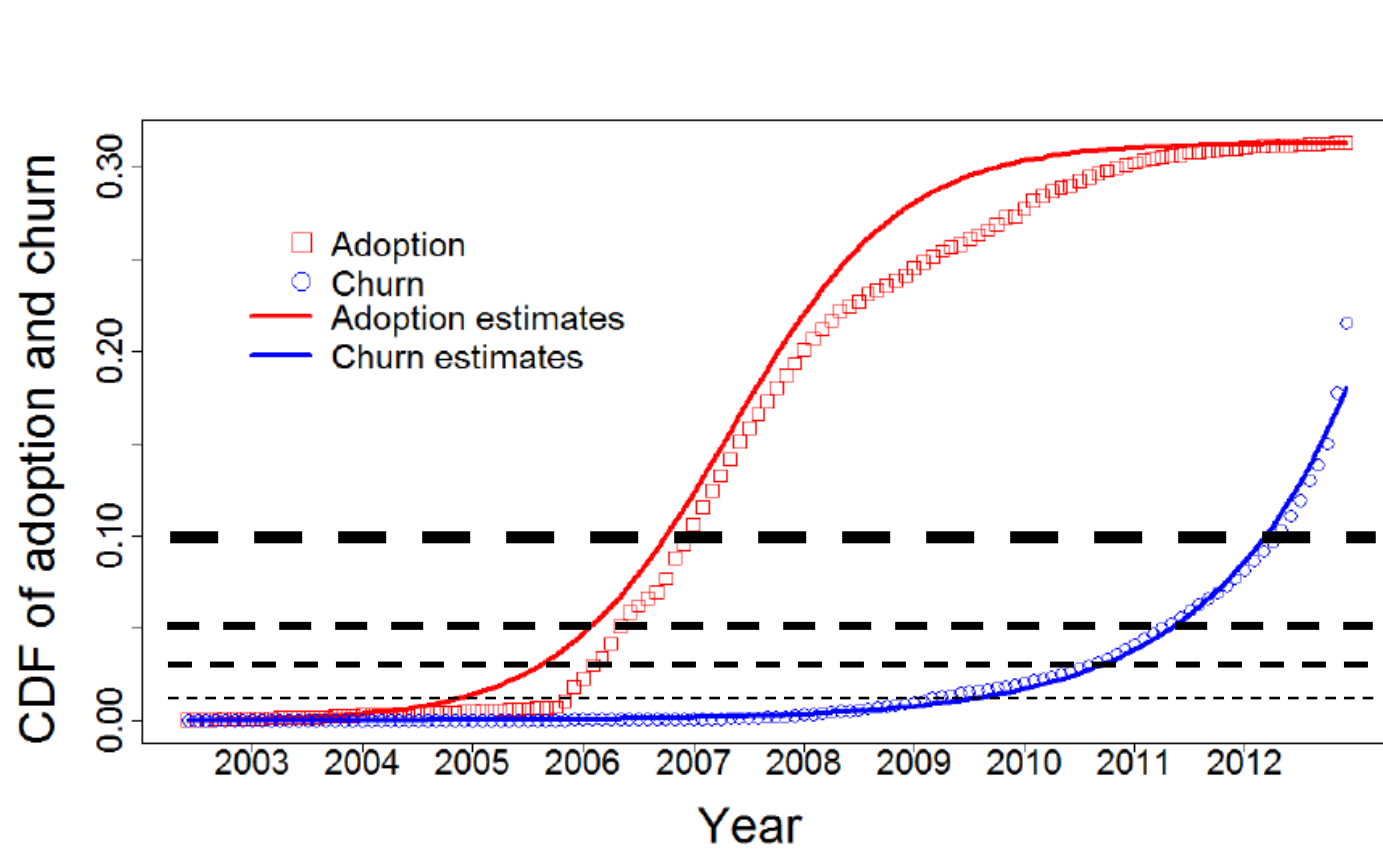
- **Location** (2557 towns),
- **Date of registration,**
- **Date of last login,**
- **Connections (time-stamp),**
- **ID of inviters**



# Those places churn early that adopt early



# Those places churn early that adopt early



# Adoption Rate

(Bass Curve)

$$\frac{dy_a(t)}{dt} = (p_a + q_a y_a(t))(1 - y_a(t))$$

# of adopters

$y_a(t)$

$t$

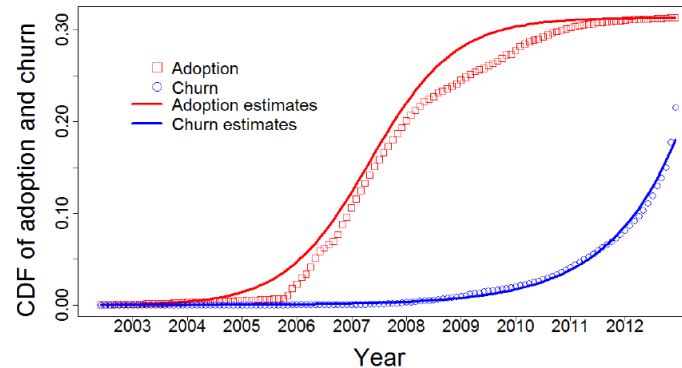
Time (months)

Innovation Par.

$p_a$

$q_a$

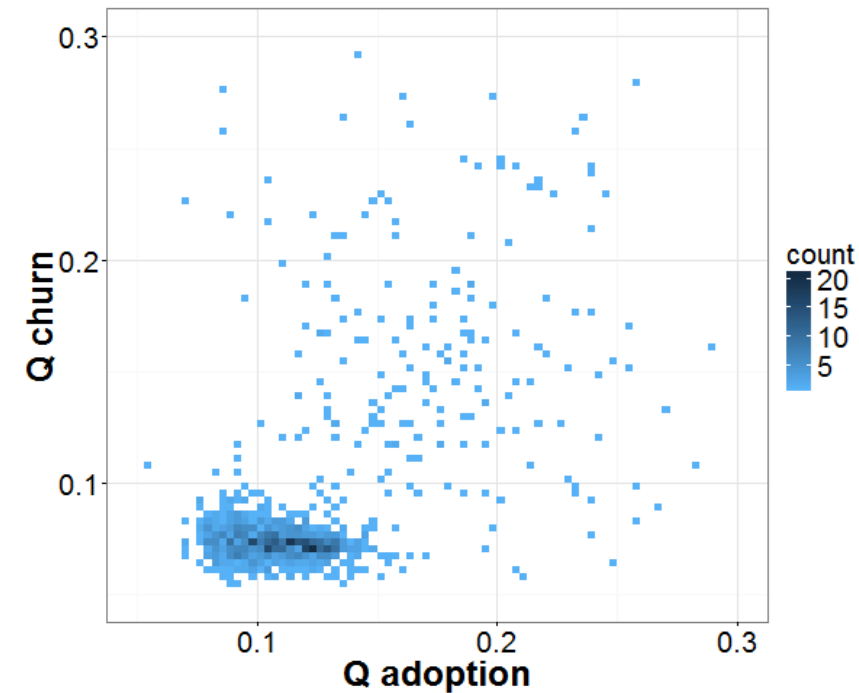
Imitation Par.



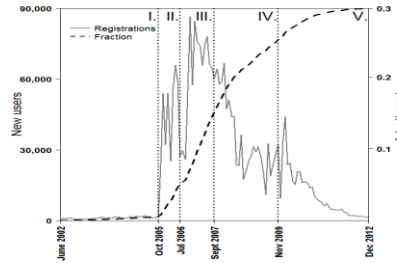
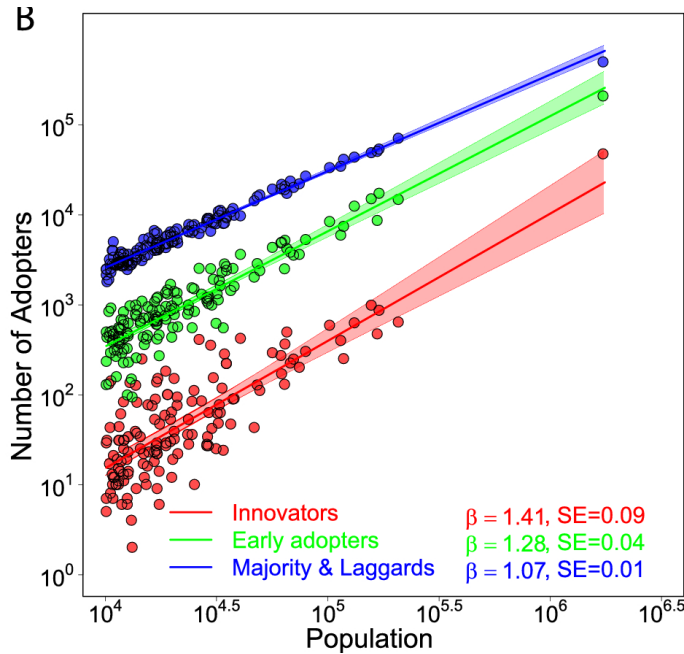
# Churn Rate

(Exponential growth)

$$\frac{dF_{ch}(t)}{dt} = q_{ch} F_{ch}(t)$$

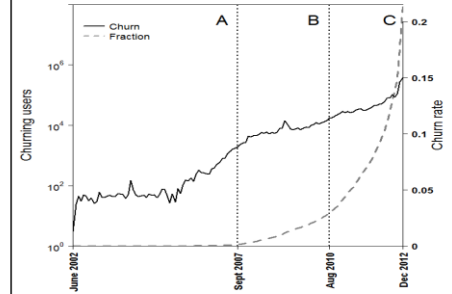
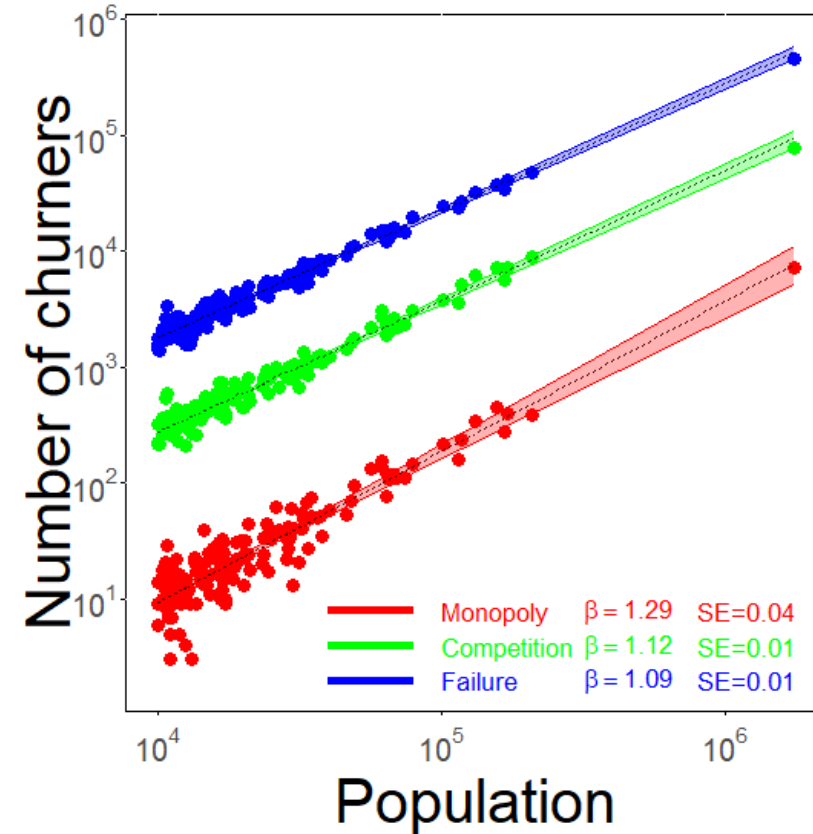


# The role of city size in adoption and churn



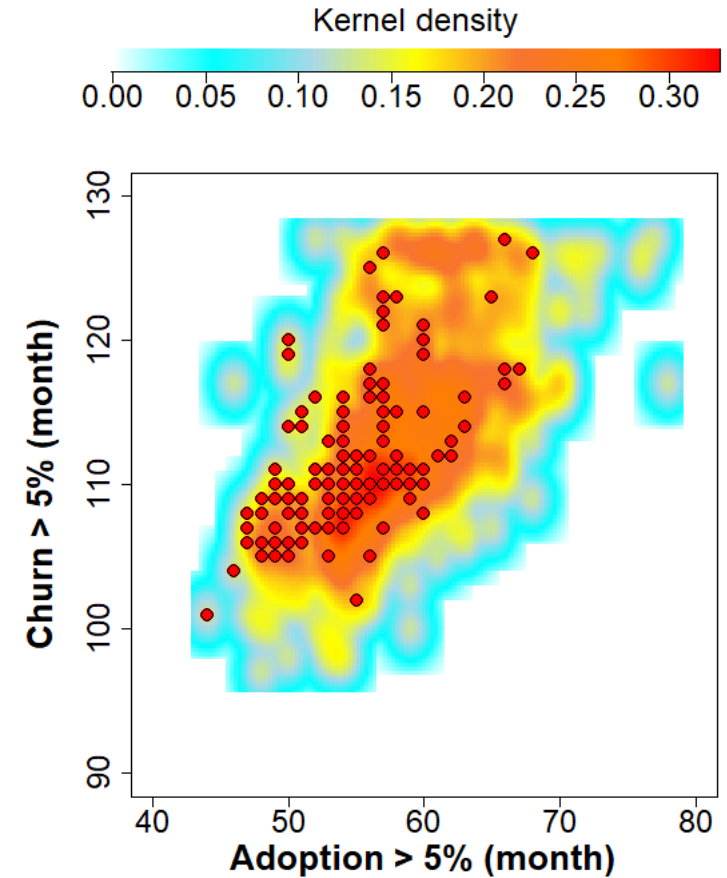
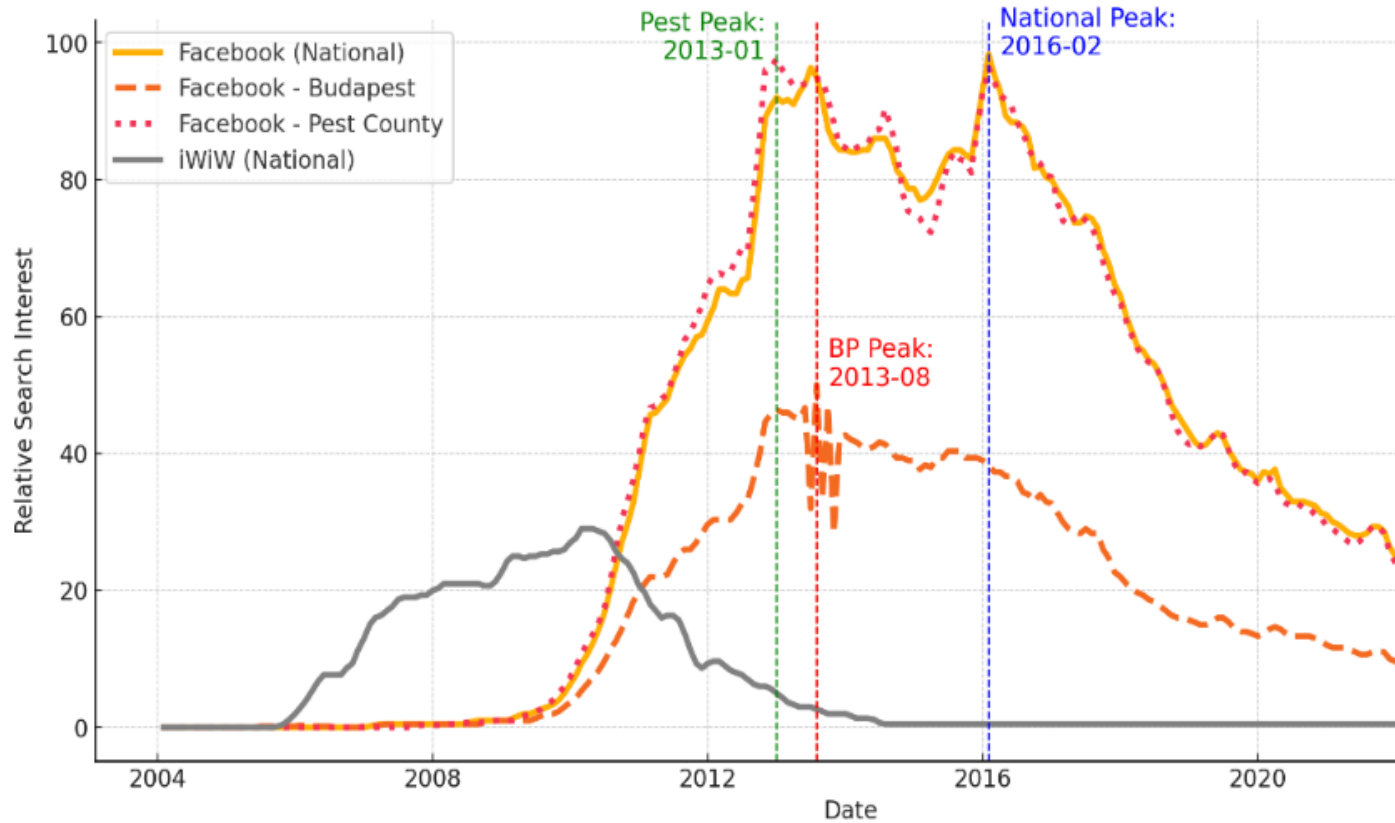
Innovators and Early Adopters concentrate in cities.

Lengyel et al. (2020, Scientific Reports)

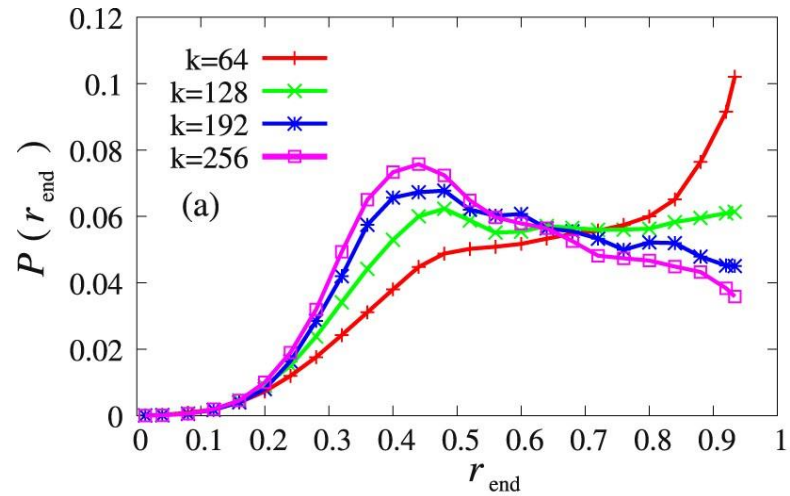




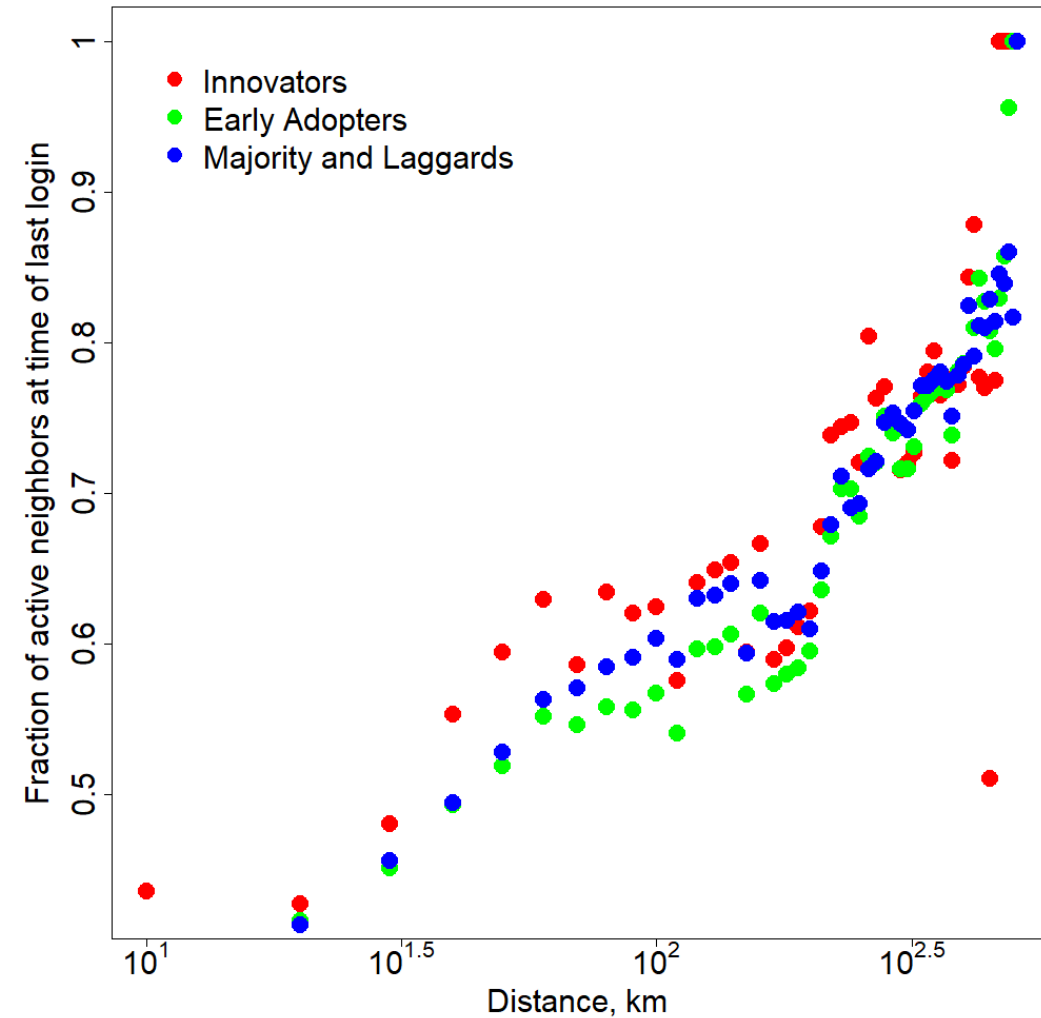
# The arrival of the new technology



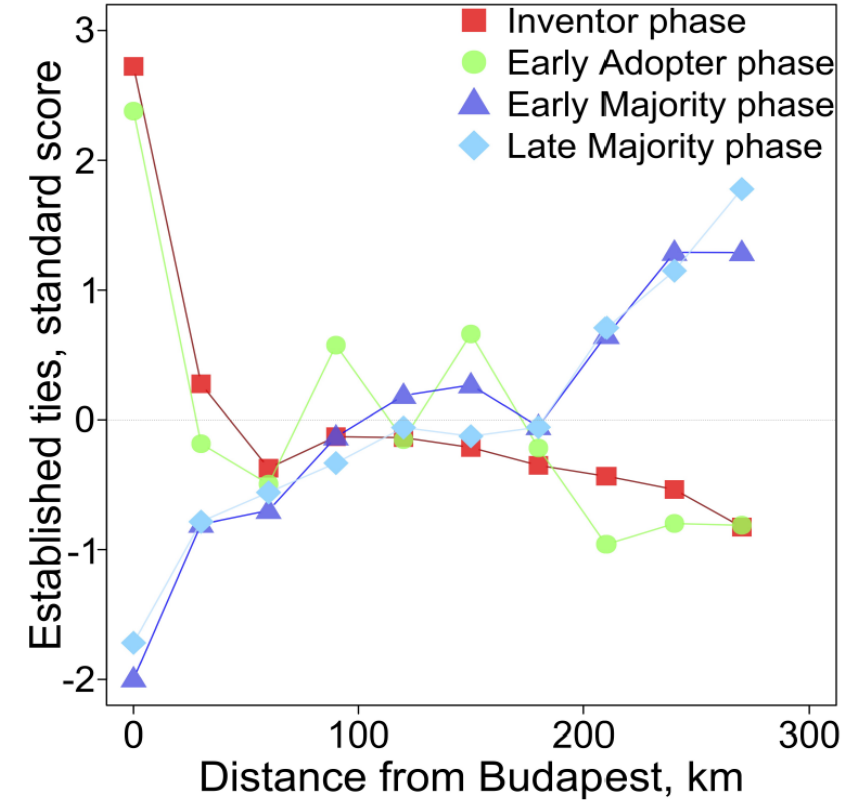
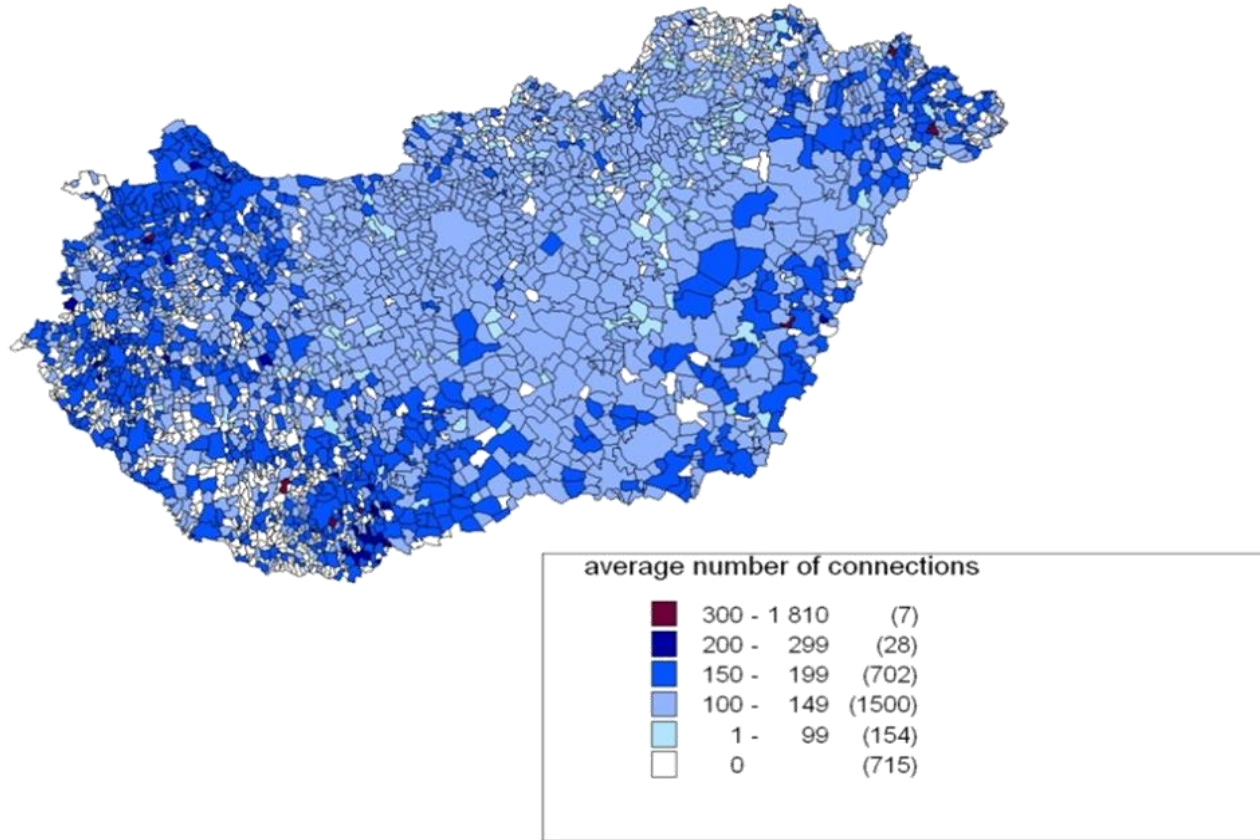
# Peer effects in churn are local



Török and Kertész (2017, Scientific Reports)



# Big waves of technology usage: reporting friends

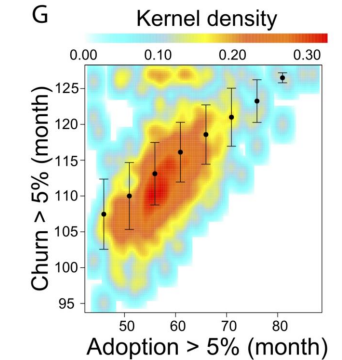


**Determinants of adoption and churn dynamics, town-level OLS regression.** Dependent variable: Month when town-level adoption and churn rates reach 5%. Control variables not reported: Average wage (numeric) , Ratio of taxpayers (numeric), Composite Telecommunication Index (numeric) developed in Lengyel and Jakobi (2016), University (equals 1 if the town hosts a university, else is 0), all measured in year 2008. Standard error in parentheses. Notations \*\*\*, \*, \* stand for statistical significance is reported at the 1%, 5% and 10% levels.

|                           | Adoption, 5% (month) |                      |                      |                      |                       |
|---------------------------|----------------------|----------------------|----------------------|----------------------|-----------------------|
|                           | (1)                  | (2)                  | (3)                  | (4)                  | (5)                   |
| Population, log           | -3.994***<br>(0.303) |                      |                      | -2.479***<br>(0.256) | -2.260***<br>(0.2543) |
| Distance from Center, log |                      | 7.791***<br>(0.484)  |                      | 2.749***<br>(0.411)  |                       |
| Central Region (dummy)    |                      |                      | -5.138***<br>(0.433) |                      | -3.160***<br>(0.332)  |
| Constant                  | 72.580***<br>(1.035) | 43.389***<br>(0.982) | 59.732***<br>(0.157) | 75.049***<br>(1.593) | 81.781***<br>(1.265)  |
| Town Controls             | NO                   | NO                   | NO                   | YES                  | YES                   |
| N                         | 1,324                | 1,322                | 1,324                | 1,322                | 1,324                 |
| R2                        | 0.116                | 0.164                | 0.096                | 0.549                | 0.564                 |
| F                         | 174.276***           | 259.242***           | 140.854***           | 266.774***           | 284.192***            |

|                           | Churn, 5% (month)    |                       |                       |                       |                       |                       |                    |
|---------------------------|----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|--------------------|
|                           | (0)                  | (1)                   | (2)                   | (3)                   | (4)                   | (5)                   | (6)                |
| Month of adoption 5%      | 0.865***<br>(0.035)  |                       |                       |                       |                       |                       | 0.984*<br>(0.050)  |
| Population, log           |                      | -7.601***<br>(0.424)  |                       |                       | -7.669***<br>(0.521)  | -7.591***<br>(0.525)  | -5.342*<br>(0.458) |
| Distance from Center, log |                      |                       | 2.945***<br>(0.717)   |                       | -2.158***<br>(0.792)  |                       | -3.461*<br>(0.676) |
| Central Region (dummy)    |                      |                       |                       | -2.793***<br>(0.612)  |                       | -0.391<br>(0.613)     |                    |
| Constant                  | 62.961***<br>(2.033) | 139.956***<br>(1.499) | 107.409***<br>(1.446) | 113.702***<br>(0.238) | 154.070***<br>(3.579) | 147.757***<br>(2.942) | 76.562*<br>(7.056) |
| Town Controls             | NO                   | NO                    | NO                    | NO                    | YES                   | YES                   | YES                |
| N                         | 1,021                | 1,021                 | 1,021                 | 1,021                 | 1,021                 | 1,021                 | 1,021              |
| R2                        | 0.377                | 0.240                 | 0.016                 | 0.020                 | 0.302                 | 0.297                 | 0.496              |
| F                         | 617.344***           | 321.966***            | 16.873***             | 20.814***             | 73.155***             | 71.490***             | 142.385            |



**Social connections and churn dynamics, town-level OLS regression.** Dependent variable: Month when town-level churn rates reach 5%. Control variables not reported: Average wage (numeric) , Ratio of taxpayers (numeric), Composite Telecommunication Index (numeric) developed in Lengyel and Jakobi (2016), University (equals 1 if the town hosts a university, else is 0), all measured in year 2008. Standard error in parentheses. Notations \*\*\*, \*, \* stand for statistical significance is reported at the 1%, 5% and 10% levels.

|                                    | <b>Churn, 5% (month)</b> |                      |                      |
|------------------------------------|--------------------------|----------------------|----------------------|
|                                    | (1)                      | (2)                  | (3)                  |
| <b>Month of adoption 5%</b>        | 0.838***<br>(0.037)      | 0.933***<br>(0.041)  | 1.169***<br>(0.054)  |
| <b>Population, log</b>             | -5.102***<br>(0.384)     | -5.306***<br>(0.412) | -5.630***<br>(0.461) |
| <b>Avg. degree</b>                 | -0.045***<br>(0.005)     |                      |                      |
| <b>Avg. degree, Early Majority</b> |                          | 0.059***<br>(0.009)  | 0.069***<br>(0.009)  |
| <b>Avg. degree, Late Majority</b>  |                          | -0.085***<br>(0.008) | -0.081***<br>(0.008) |
| <b>Avg. Degree, Laggards</b>       |                          | 0.000***<br>(0.000)  | 0.000***<br>(0.000)  |
| <b>Constant</b>                    | 91.058***<br>(2.831)     | 81.273***<br>(3.384) | 58.084***<br>(5.401) |
| <b>Town Controls</b>               | NO                       | NO                   | YES                  |
| <b>N</b>                           | 1,021                    | 1,021                | 1,021                |
| <b>R2</b>                          | 0.492                    | 0.512                | 0.535                |
| <b>F</b>                           | 328.188***               | 213.014***           | 129.173***           |



# Further steps: generalization and synergies

## **Different contexts**

- Web technologies
- Technologies from patent (and trademark) data
- Scientific terms, concepts, keywords in publications
- Topics and expressions in news

## **Mechanisms**

- Better identification of substituting technologies

## **Theory and models**

- Recovery like in the network epidemiology models
- Regional economics and economic development

Thank you!



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